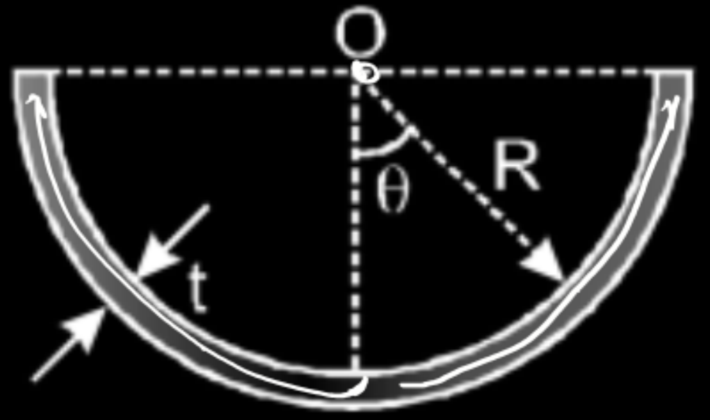
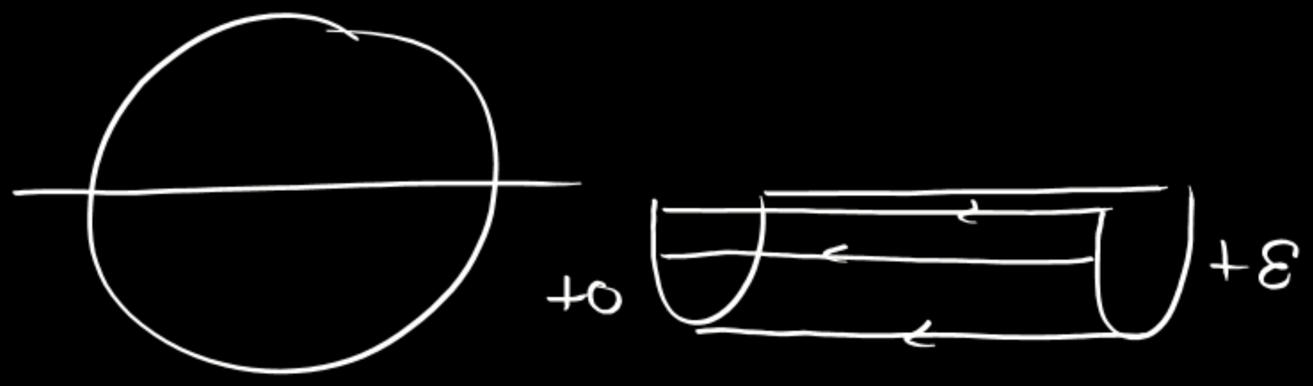
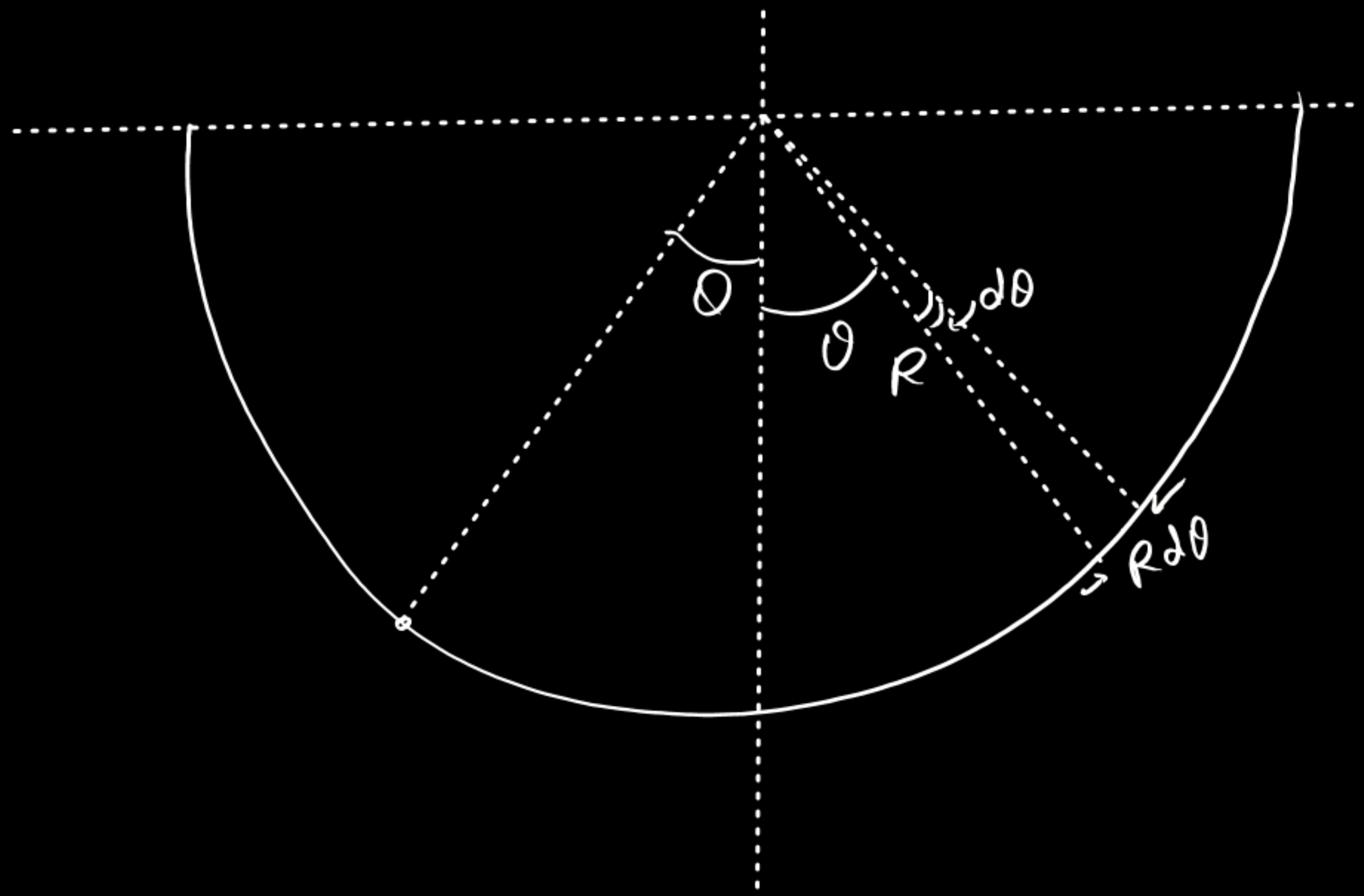
Conductor of length ℓ has shape of a semi cylinder of radius R ($<< \ell$). Cross section of the conductor is shown in the figure. Thickness of the conductor is t ($<<$ R) and conductivity of its material varies with angle θ according to the law $\sigma = \sigma_0 \cos\theta$ where σ_0 is a constant. If a battery of emf ε is connected across its end faces (across the semi–circular cross-sections), the magnetic induction at the mid point O of the axis of the semi-cylinder is :

$\sigma = \sigma_0 \cos\theta$





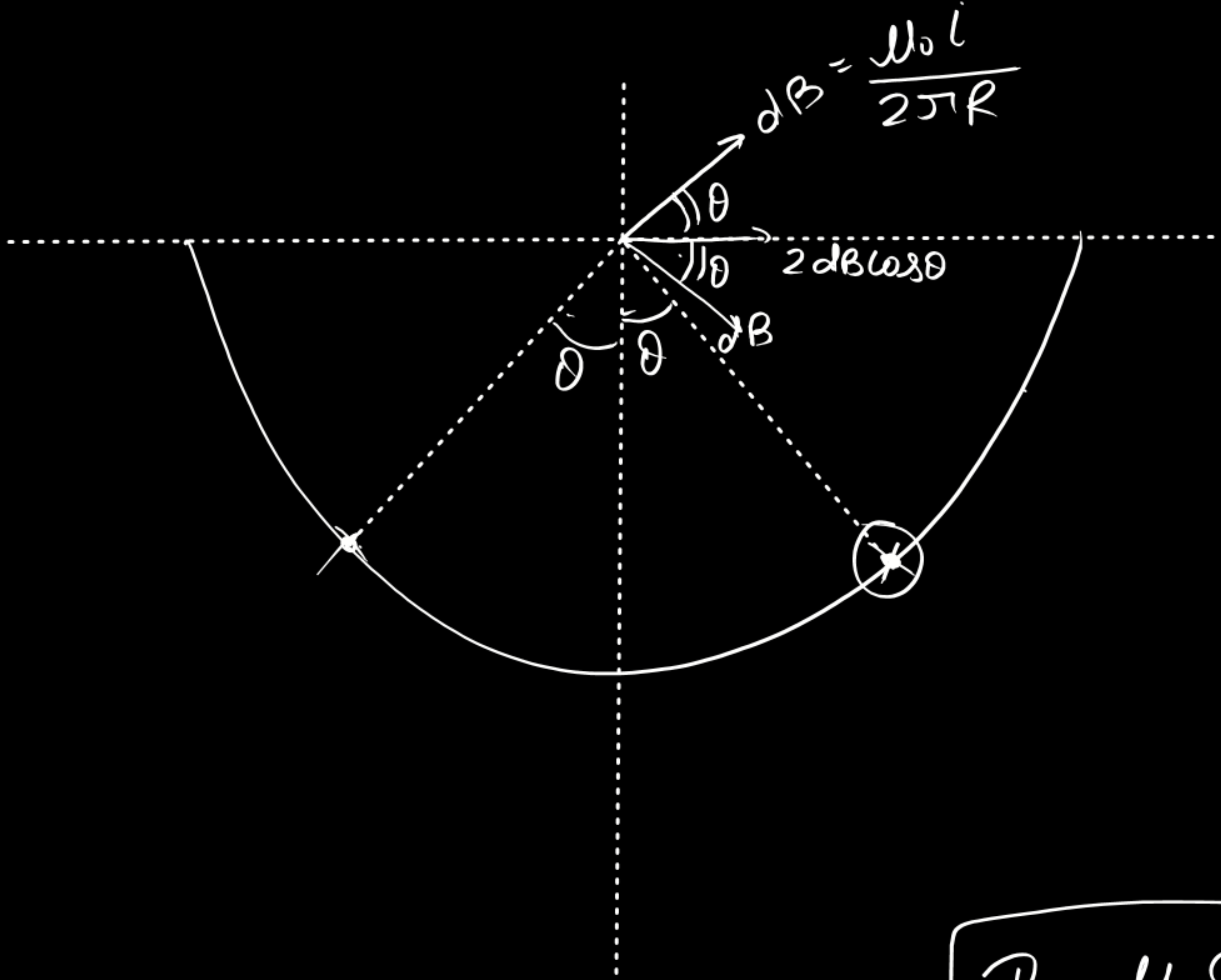
(A) $\frac{\mu_0 \sigma_0 \varepsilon t}{8\ell}$ (B) $\frac{\mu_0 \sigma_0 \varepsilon t}{4\ell}$ (C) $\frac{\mu_0 \sigma_0 \varepsilon t}{\ell}$ (D) $\frac{2\mu_0 \sigma_0 \varepsilon t}{\ell}$



Resistance

$$= \frac{l}{\sigma_0 \cos\theta \, R t \, d\theta}$$

$$\boxed{i = \frac{\varepsilon \sigma_0 R t \cos\theta \, d\theta}{l}}$$



$$B_{net} = \int 2\,dB\cos\theta$$

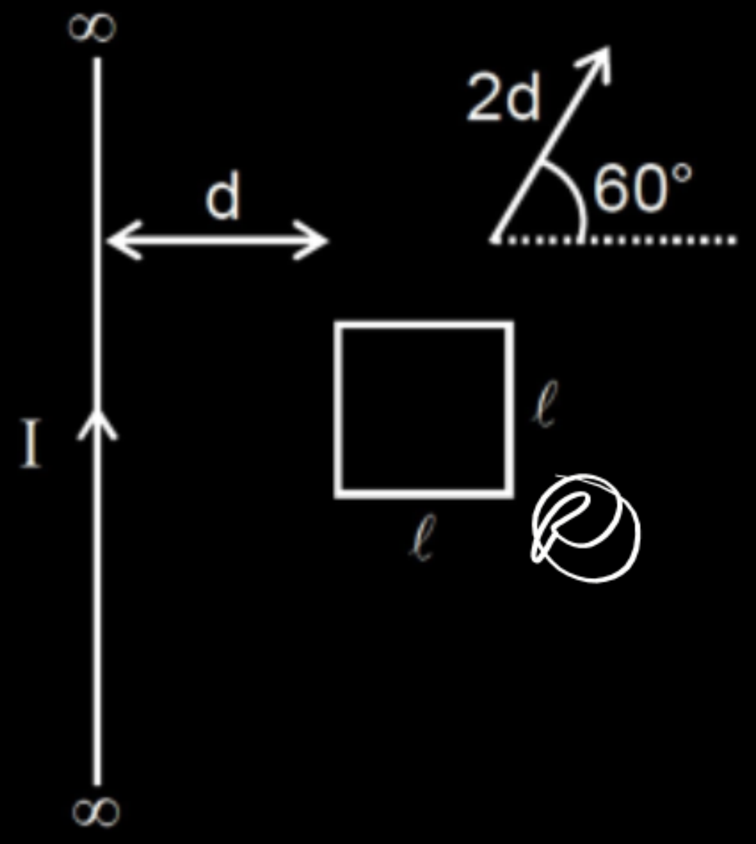
$$= \cancel{2}\int_0^{\pi/2} \frac{\mu_0 \varepsilon \sigma_0 R t}{\cancel{2}\pi R\, l}\cos^2\theta\, d\theta$$

$$= \frac{\mu_0 \varepsilon \sigma_0 t}{\pi l}\left(\frac{\theta}{2} + \frac{\sin 2\theta}{2}\right)_0^{\pi/2}$$

$$= \frac{\mu_0 \varepsilon \sigma_0 t}{\cancel{\pi} l}\cdot\frac{\cancel{\pi}}{4}$$

$$\boxed{B = \frac{\mu_0 \varepsilon \sigma_0 t}{4l}}$$

A square loop of side length ℓ is placed near an infinitely long straight current carrying wire. Current in the wire is I. Distance of the nearer side of the loop from the wire is d. The wire and the loop are coplanar. Total resistance of the loop is R. If the loop translates through distance 2d in a direction shown in the figure, with the loop always remaining coplanar with wire then the charge flowing in it during the given displacement will be :



$$\boxed{Q = \frac{\Delta\phi}{R}}$$

(A) $\frac{\mu_0 I\ell}{2\pi R}\ell n\left(\frac{d+\ell}{d}\right)$ (B) $\frac{\mu_0 I\ell}{2\pi R}\ell n\left(\frac{2d+\ell}{2d}\right)$ (C) $\frac{\mu_0 I\ell}{2\pi R}\ell n\left(\frac{d+\ell}{2d+\ell}\right)$ (D) $\frac{\mu_0 I\ell}{2\pi R}\ell n\left(\frac{2d+2\ell}{2d+\ell}\right)$

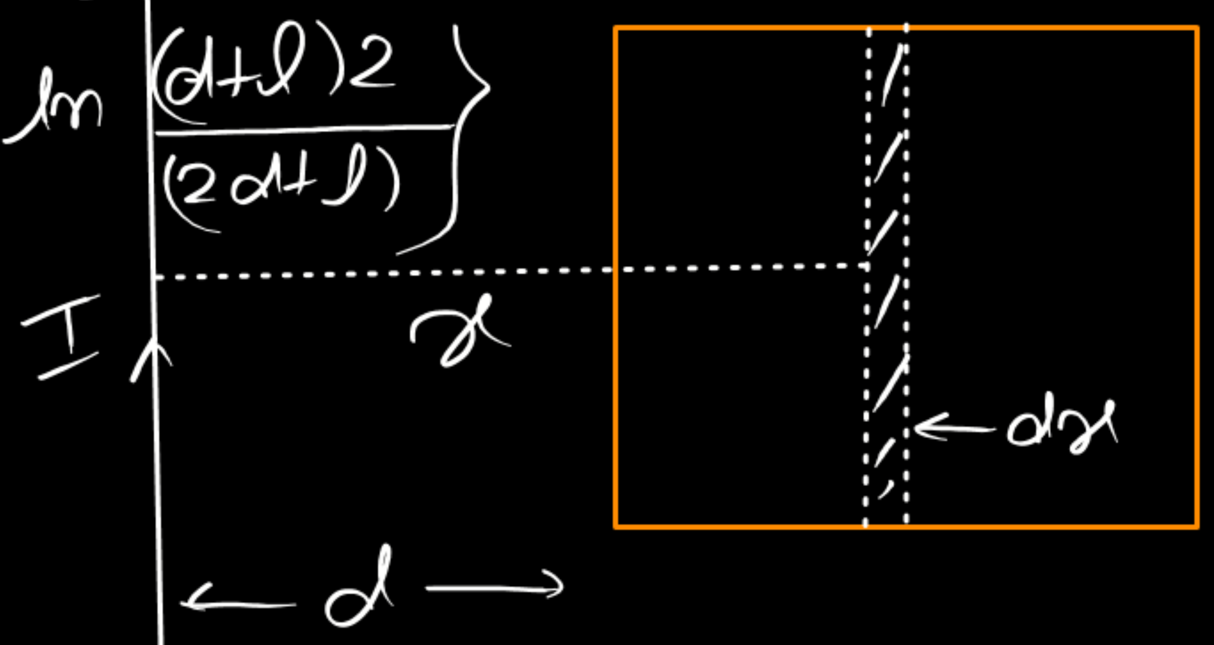
$$Q = \frac{\Delta \Phi}{R}$$

$$= \frac{\mu_0 I l}{2\pi R}\left\{ \ln \frac{d+l}{d} - \ln \frac{2d+l}{2d} \right\} l$$

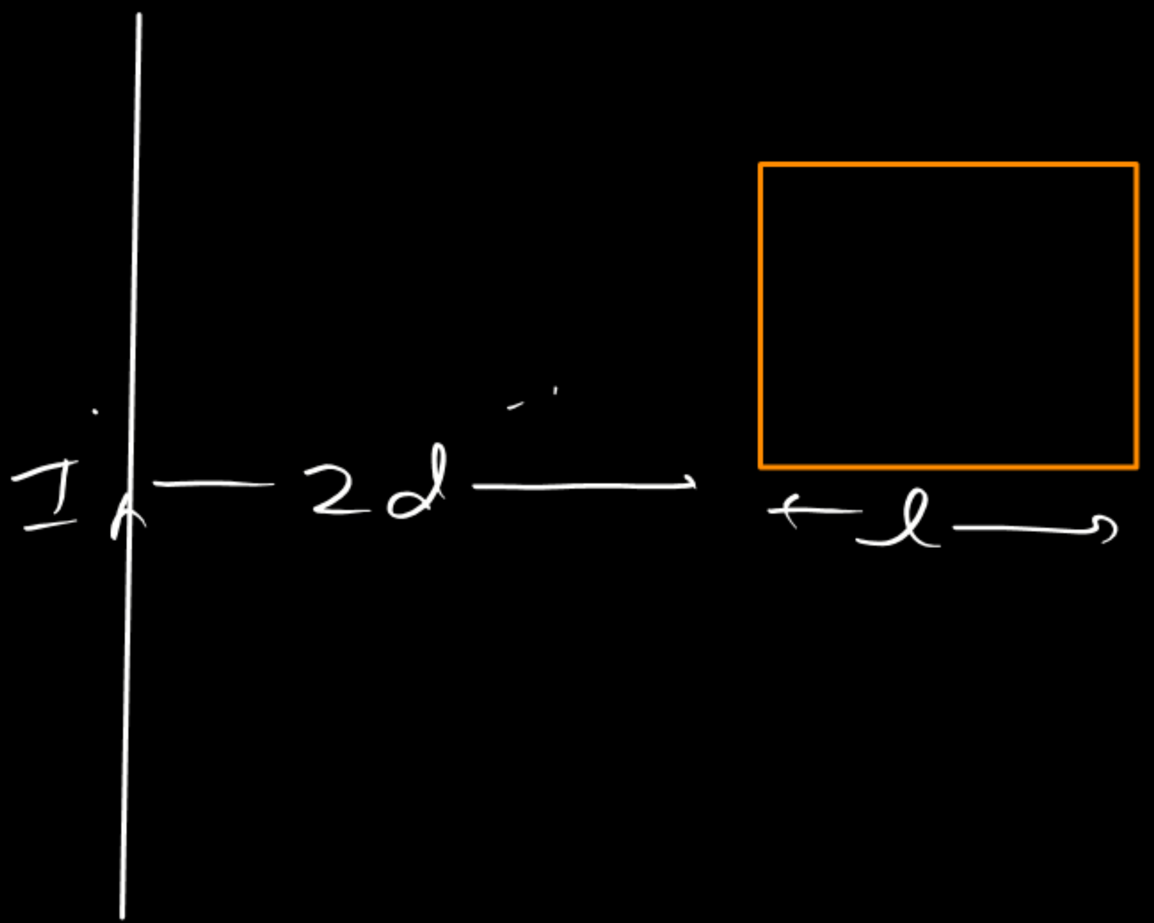
$$Q = \frac{\mu_0 I l}{2\pi R}\left\{ \ln \frac{(d+l)2}{(2d+l)} \right\}$$

$$\Phi = \int_{d}^{d+l} \frac{\mu_0 I}{2\pi x}\, l\, dx \cos 0$$

$$\Phi_i = \frac{\mu_0 I l}{2\pi} \ln \frac{d+l}{d}$$

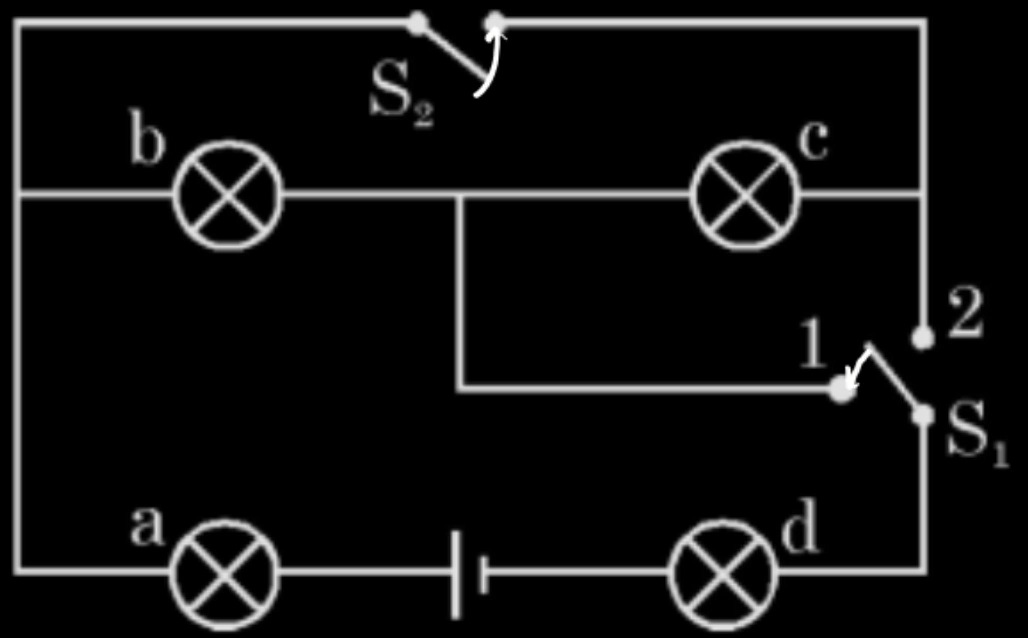


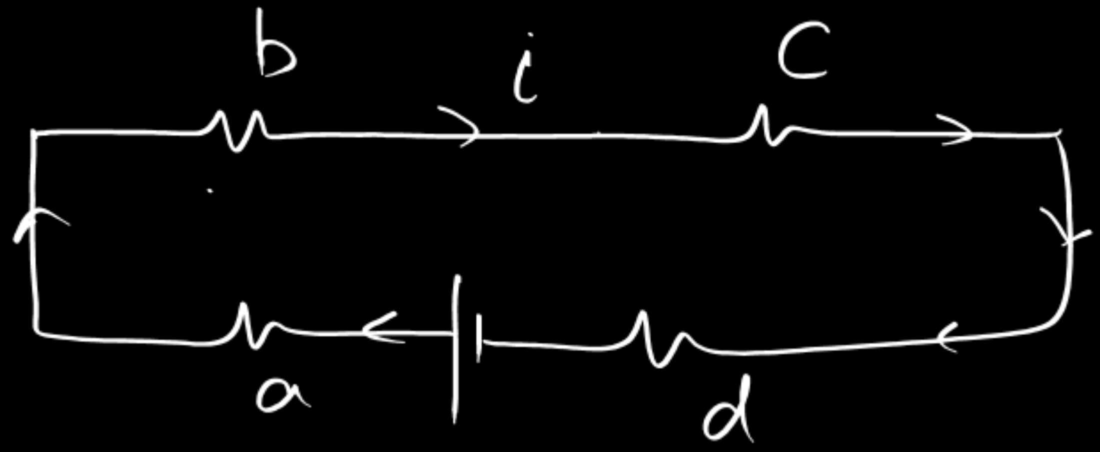
$$\Phi_f = \frac{\mu_0 I l}{2\pi} \ln \frac{2d+l}{2d}$$



Four lamps are connected in the way shown in the figure. When switch S_2 is open and switch S_1 is on position-2, Lamp-b is the brightest, and lamp-c and lamp-d are the dimmest and are of the same brightness. Now S_2 is closed and S_1 is on position–1, the correct order of brightness of the lamps with the first in the sequence being the brightest is :

$b > a > c = d$





$P = \frac{V^2}{R} = i^2R$

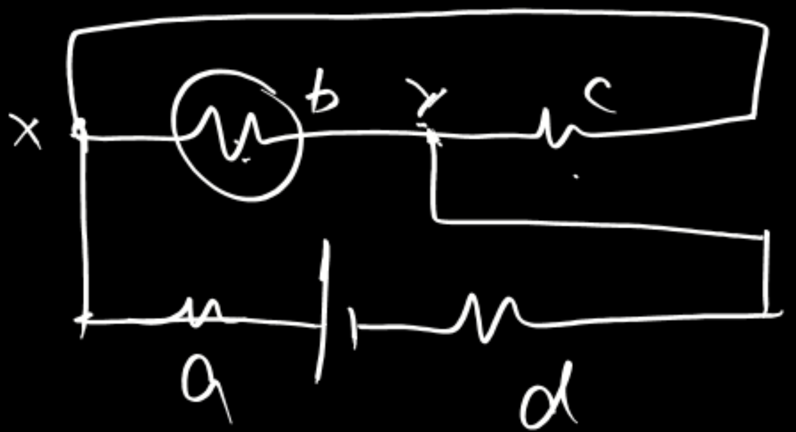
(A) c, d, b, a

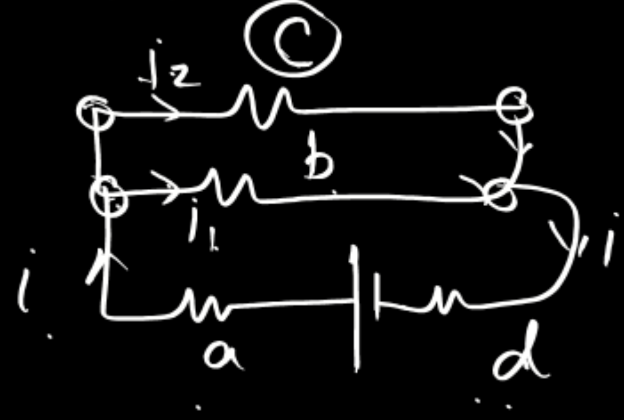
(B) a, d, b, c

(C) a, b, c, d

(D) a, d, c, b

$P = \frac{V^2}{R}$

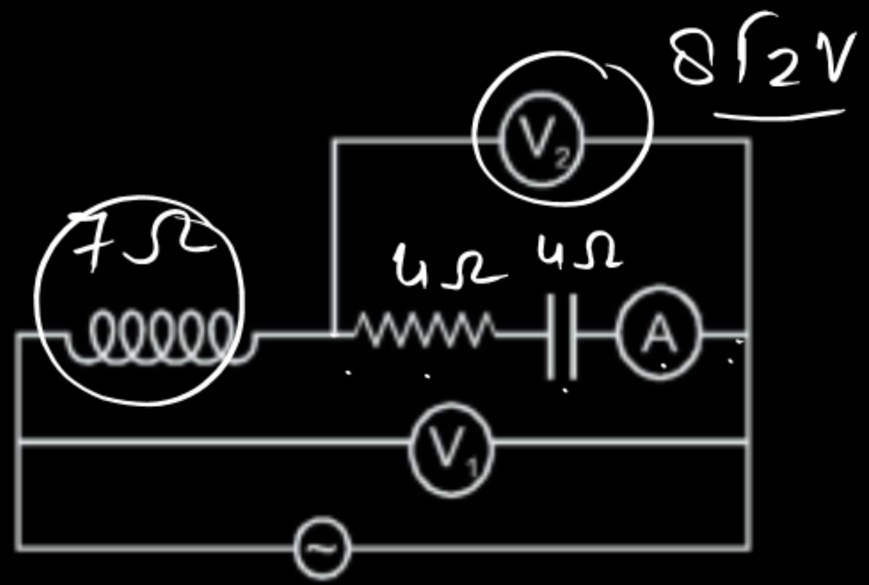




$a > d > c > b$

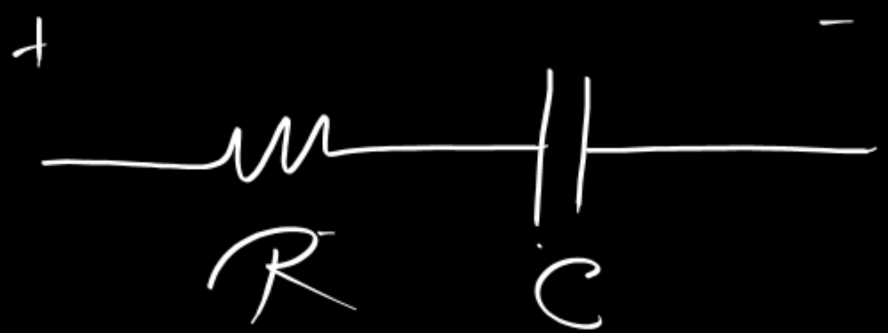
In the ac circuit shown, $X_L = 7\ \Omega$, $R = 4\ \Omega$ and $X_C = 4\ \Omega$. The reading of the ideal voltmeter V_2 is $8\sqrt{2}$.

The reading of the ideal ammeter :



(A) 3A (B) 2A (C) 4A (D) 5A

$8\sqrt{2} = i_{Rms} Z$



$Z = \sqrt{4^2 + 4^2}$

$= 4\sqrt{2}$

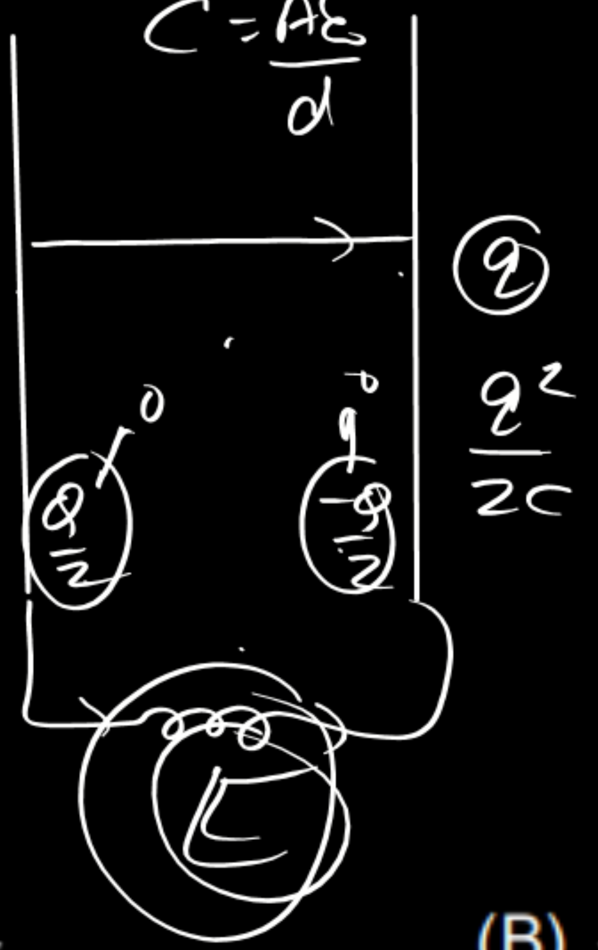
$i = \dfrac{8\sqrt{2}}{4\sqrt{2}} = 2\ Amp$

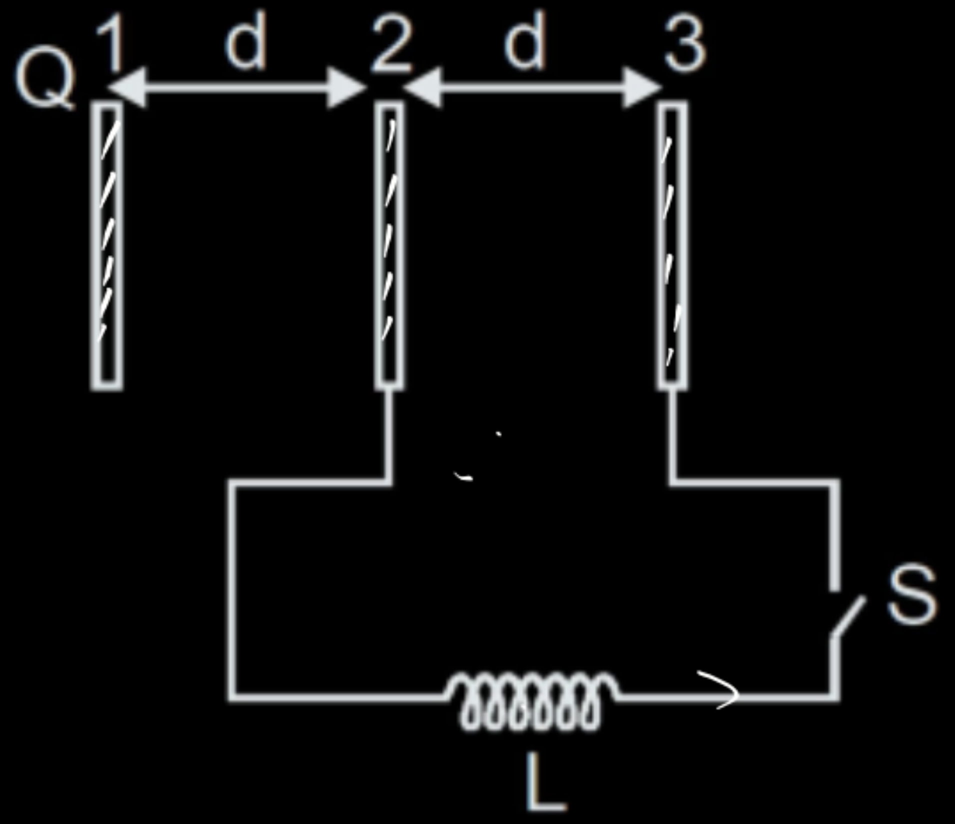
Three identical large plates are fixed at separation of d from each other as shown. The area of eachplate is A. Plate 1 is given charge +Q while plates 2 and 3 are neutral and are connected to each other through coil of inductances L and switch S. If resistance of all connected wires is neglected the maximum current flow through coil after closing switch is ($C = \varepsilon_0 A/d$) (neglect fringe effect)

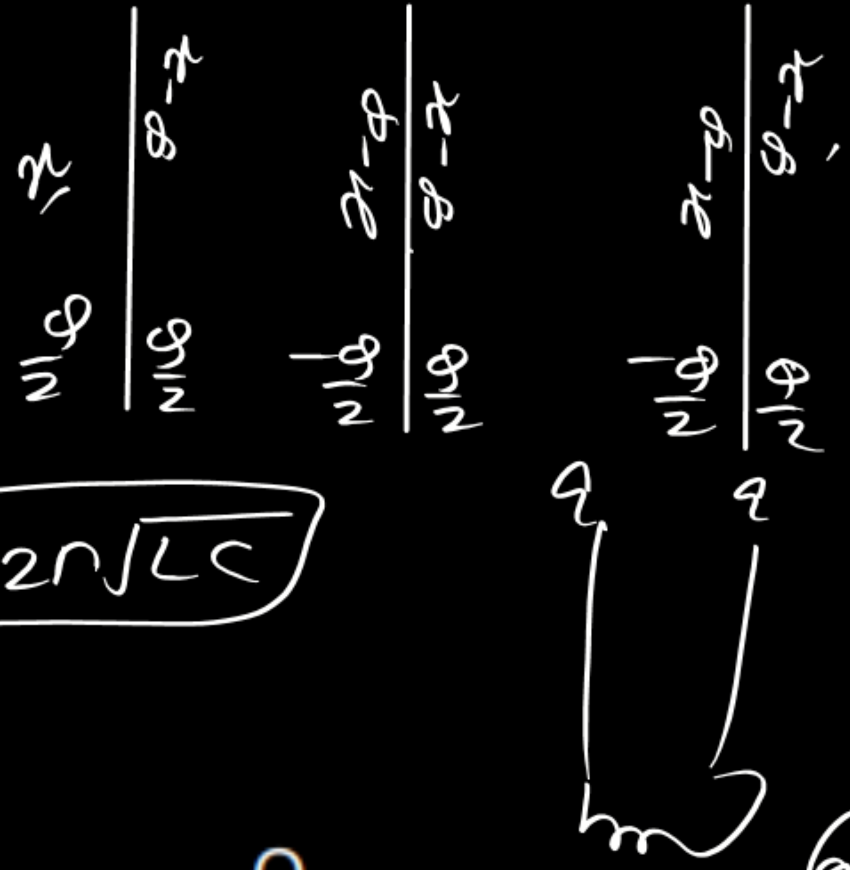
$x = q - x$

$x = \frac{q}{2}$

$C = \frac{A\varepsilon}{d}$







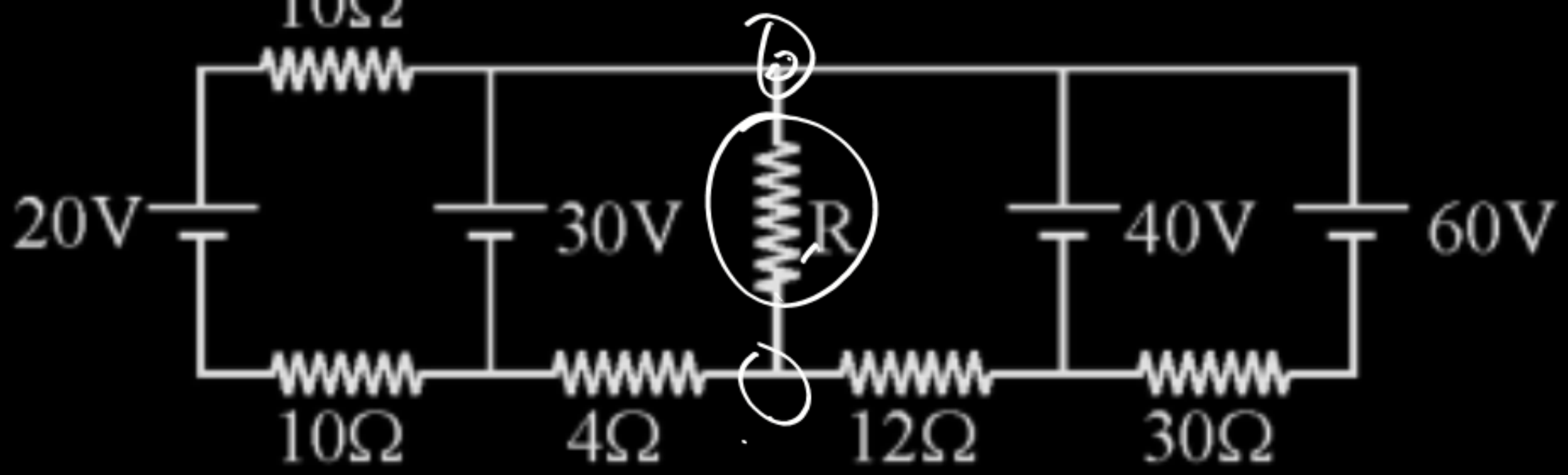
$\boxed{T = 2\pi\sqrt{LC}}$

(A) $\frac{Q_0}{\sqrt{LC}}$ (B) $\frac{Q_0}{\sqrt{2LC}}$ (C) $\frac{2Q_0}{\sqrt{LC}}$ (D) $\frac{Q_0}{2\sqrt{LC}}$

$\frac{\left(\frac{q}{2}\right)^2}{2C} = \frac{1}{2}Li^2$ $Li^2 = \frac{q^2}{4C}$ $\boxed{i = \frac{q}{2\sqrt{LC}}}$

$i = \frac{q}{\sqrt{LC}}$ $q/2$

In the given circuit, the value of R so that thermal power generated in R will be maximum .

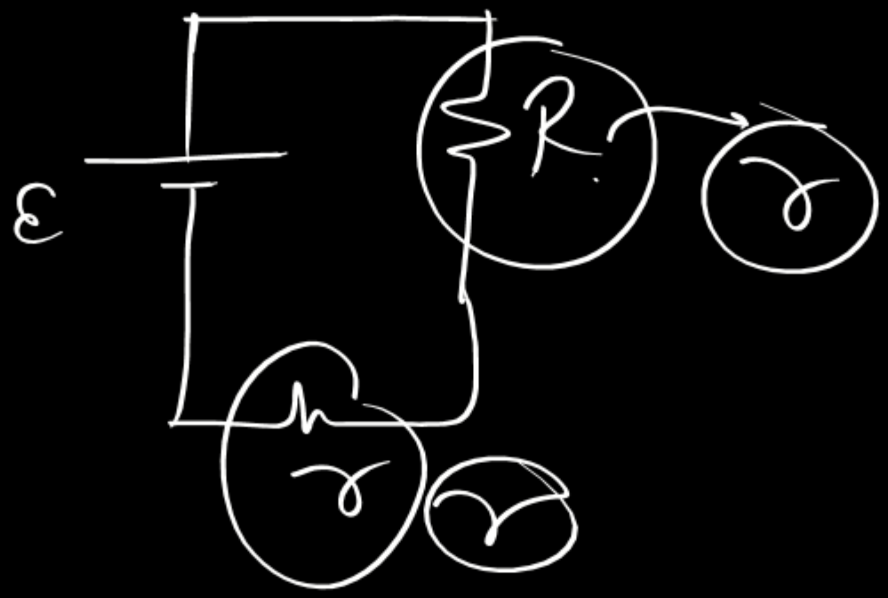


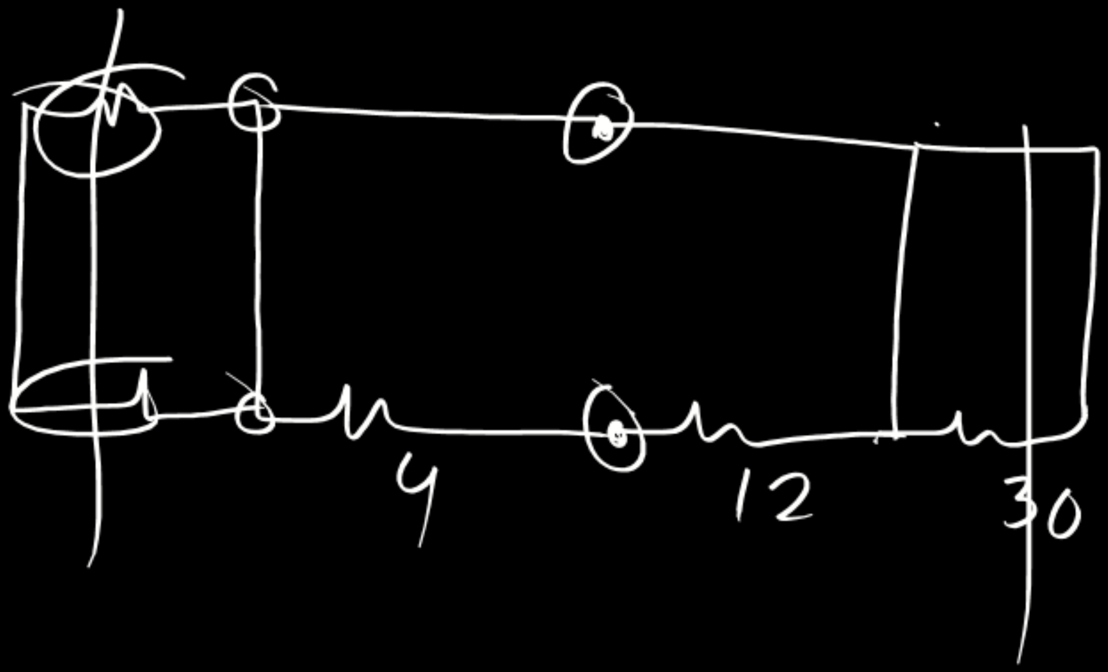
(A) 3 Ω

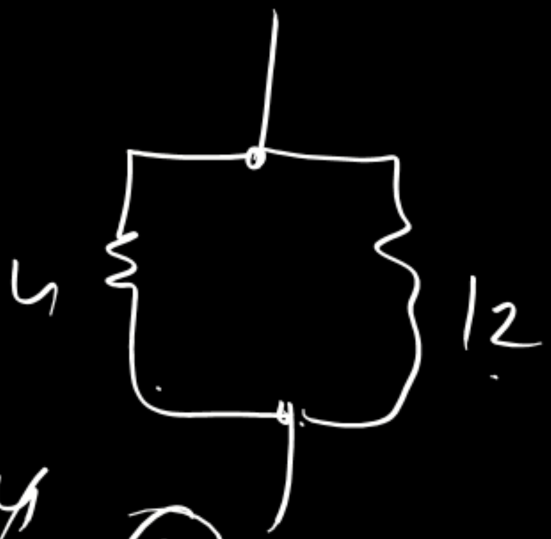
(B) 4 Ω

(C) 5 Ω

(D) 6 Ω







$\frac{12 \times 4}{16} = 3$

Consider a series LCR circuit connected to an AC supply of 220 V. If voltage drop across resistance R is V_R, voltage drop across capacitor is $V_C = 2V_R$ and that across inductor coil is $V_L = 3V_R$ then choose correct alternative(s)

(A) $V_R = 220\sqrt{2}$ V

(B) Power factor of circuit is $\frac{1}{\sqrt{2}}$

(C) $V_R = 156$ V

(D) Phase difference between current and source voltage is $\frac{\pi}{4}$

$V_R = iR$

$= \frac{220}{\sqrt{2}R} \cdot R = \frac{220}{\sqrt{2}}$

$\cos\theta = \cos 45$

$\cancel{100\sqrt{2}} = \frac{220}{\sqrt{2}} = 110 \times 1.414$

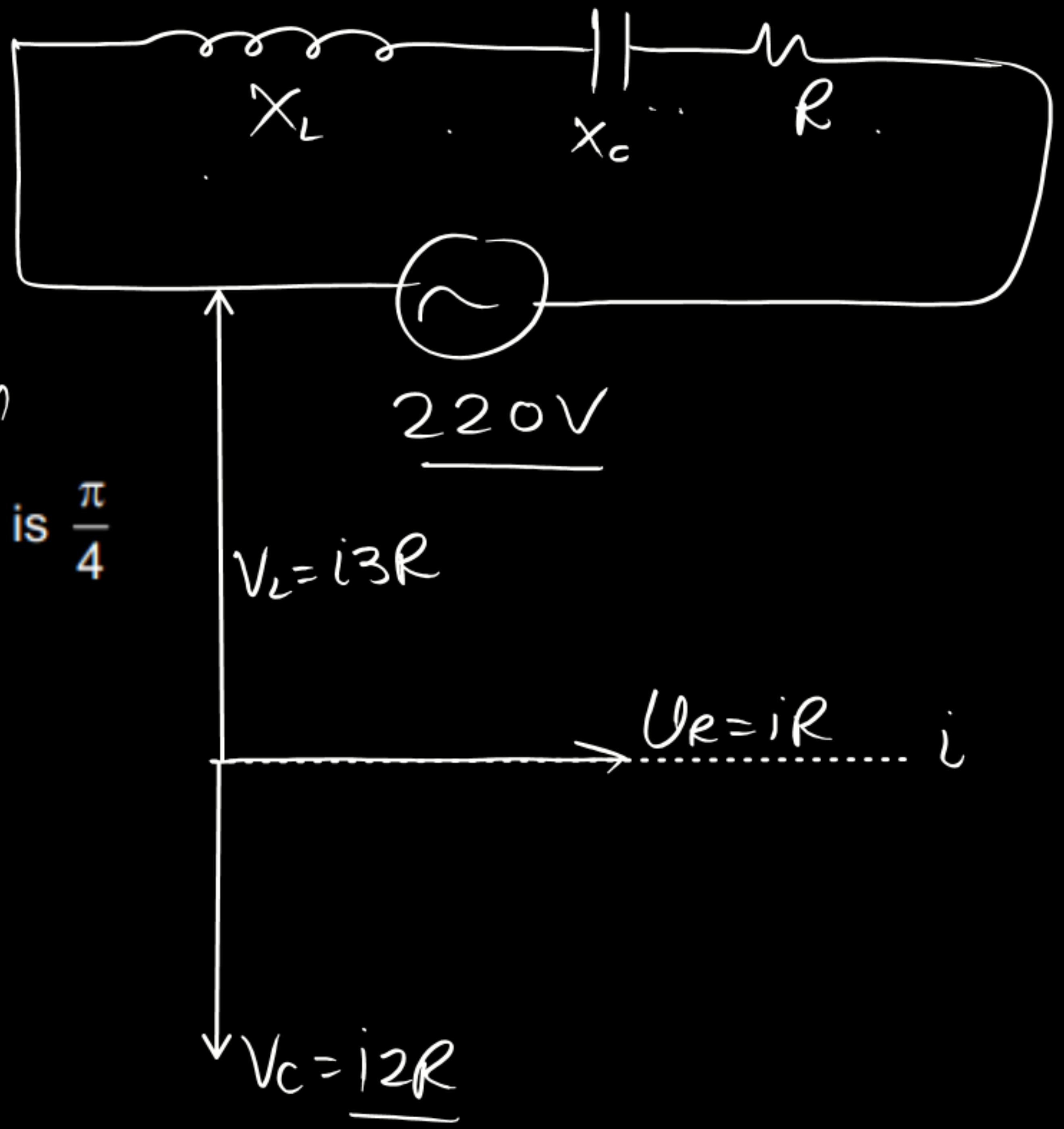
$V_R = iR$

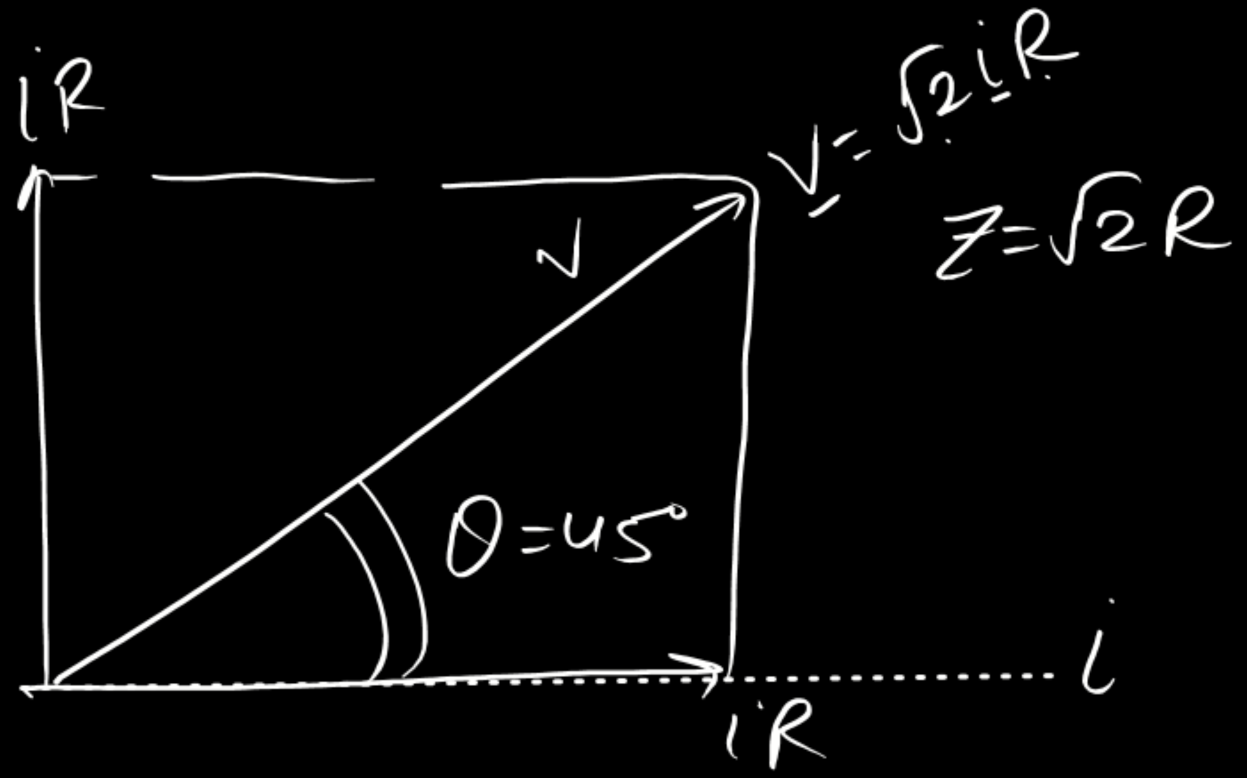
$V_C = iX_C = 2V_R =$

$\boxed{X_C = 2R}$

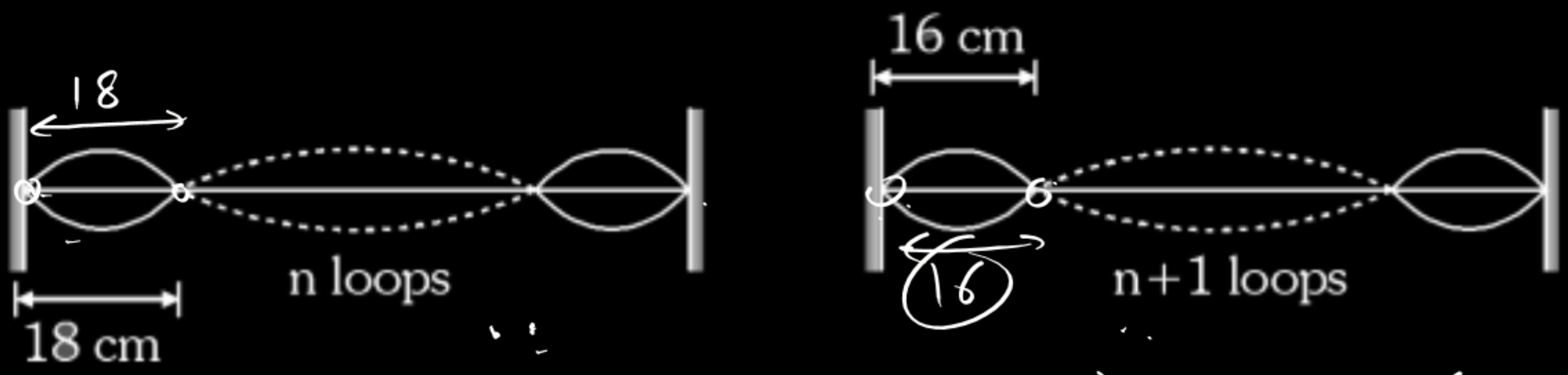
$V_L = iX_L = 3V_R$

$X_L = 3R$




iR
V = √2 iR
Z = √2 R
V
θ = 45°
i
iR

A string fixed at both ends has consecutive standing wave modes for which the distances between adjacent nodes are 18 cm and 16 cm respectively.



$$f = \frac{v}{2l} = \sqrt{\frac{T}{\mu}} \times \frac{1}{2l} = \sqrt{\frac{10}{4\times10^{-3}}} \times \frac{1}{2\times1.44} = 17.3 \text{ Hz}$$

$$l = n\times18 = (n+1)\times16$$

$$\frac{n+1}{n} = \frac{9}{8} \Rightarrow \boxed{n=8}$$

Choose the correct statement(s) :

(A) The minimum possible length of the string is 144 cm.

(B) The minimum possible length of the string is 288 cm.

(C) If the tension is 10 N and the linear mass density is 4g/m, the fundamental frequency is (17 ± 0.5) Hz

(D) If the tension is 10 N and the linear mass density is 4g/m, the fundamental frequency is (34 ± 0.5) Hz

$$l = 18\times8 = 16\times9 = \underline{144\text{cm}}$$

A man in a lift ascending with an upward acceleration 'a' throws a ball vertically upwards with a velocity ‘v’ with respect to himself and catches it after ‘t_1’ seconds. Afterwards when the lift is descending with the same acceleration 'a' acting downwards the man again throws the ball vertically upwards with the same velocity with respect to him and catches it after ‘t_2’ seconds?

(A) the acceleration of the ball w.r.t. ground is g when it is in air

(B) the velocity v of the ball relative to the lift is $\dfrac{g(t_1+t_2)}{t_1\,t_2}$ $\quad \dfrac{LT^{-2}\cancel{T}}{T\cancel{T}}$ $\quad LT^{-3}$

(C) the acceleration 'a' of the lift is $\dfrac{g\,(t_2-t_1)}{t_1+t_2}$

(D) the velocity ‘v’ of the ball relative to the man is $\dfrac{g\,t_1\,t_2}{(t_1+t_2)}$

$$t_1 = \frac{2v}{g+a} \quad \text{—}(1)$$

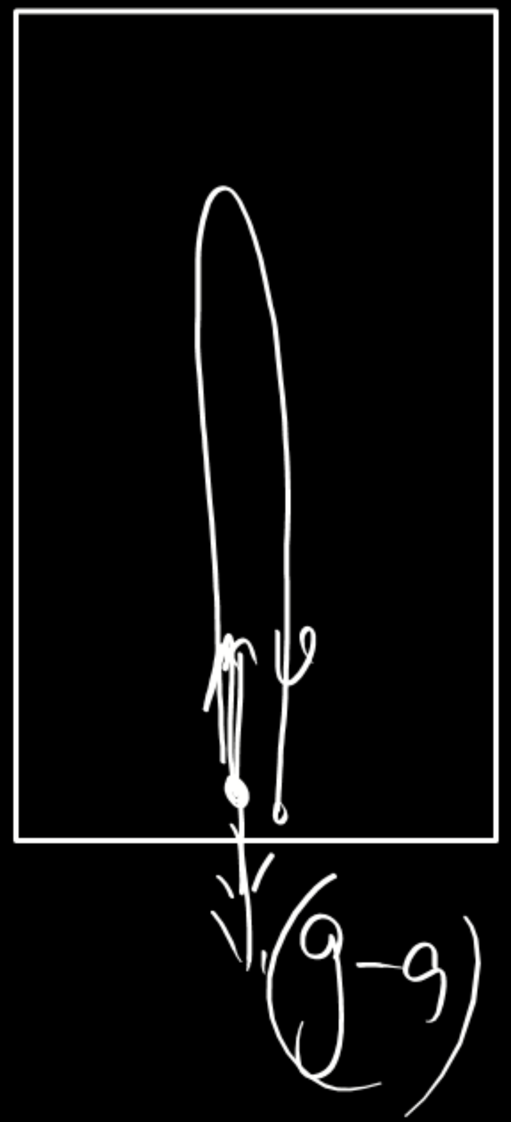
$$t_2 = \frac{2v}{g-a} \quad \text{—}(2)$$

$$\frac{t_1}{t_2} = \frac{g-a}{g+a}$$

$\downarrow a$

$$t_1 g + t_1 a = g t_2 - a t_2$$

$$a = \frac{g(t_2-t_1)}{t_1+t_2}$$



$$g+a = \frac{2v}{t_1} \qquad \cancel{2}g = \cancel{2}v\left\{\frac{1}{t_1}+\frac{1}{t_2}\right\}$$

$$g-a = \frac{2v}{t_2} \qquad v = \frac{g t_1 t_2}{t_1+t_2}$$

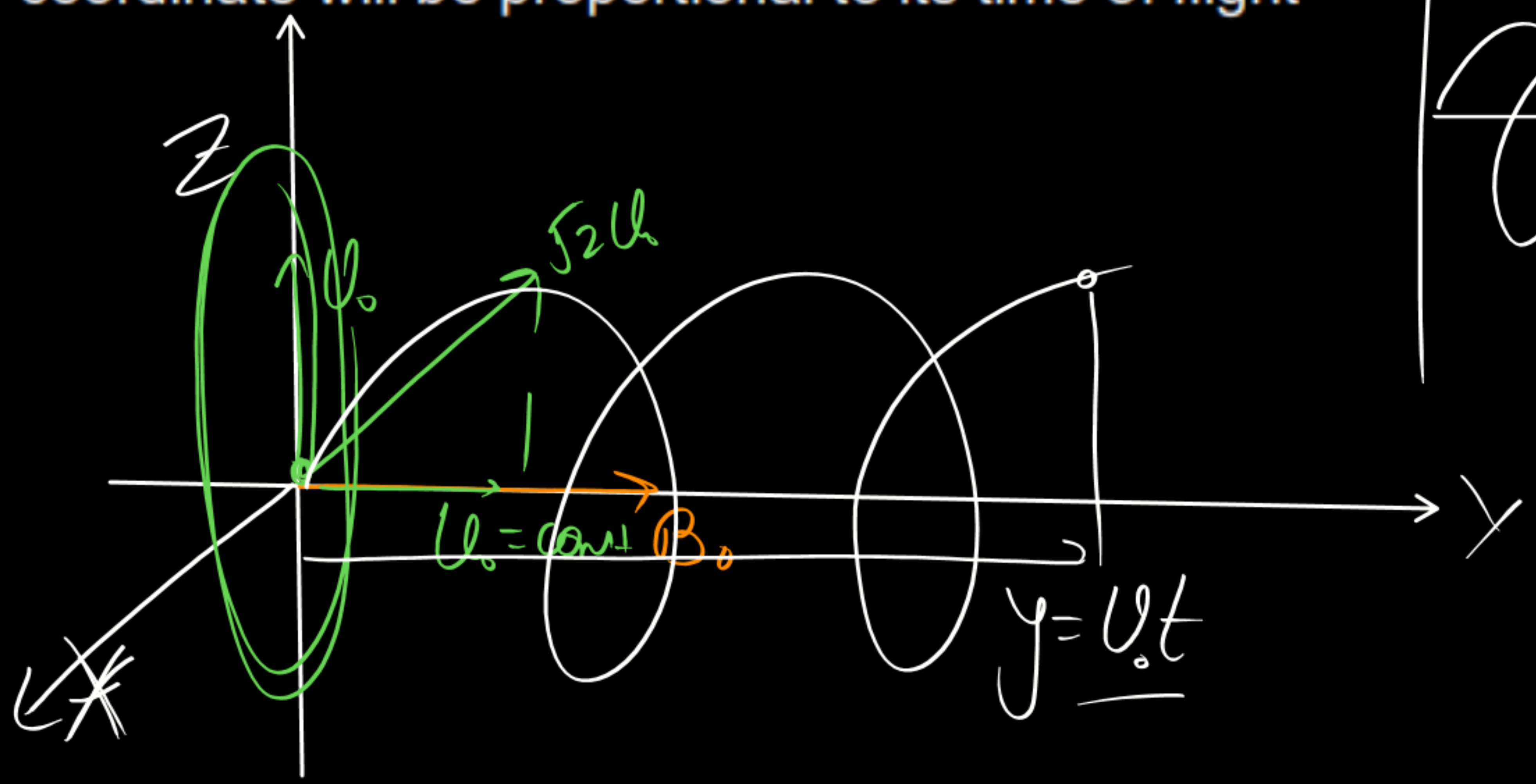
A proton is fired from origin with velocity $\vec{v} = v_0\hat{j} + v_0\hat{k}$ in a uniform magnetic field $\vec{B} = B_0\hat{j}$ In the subsequent motion of the proton (v_0 and B_0 are positive)

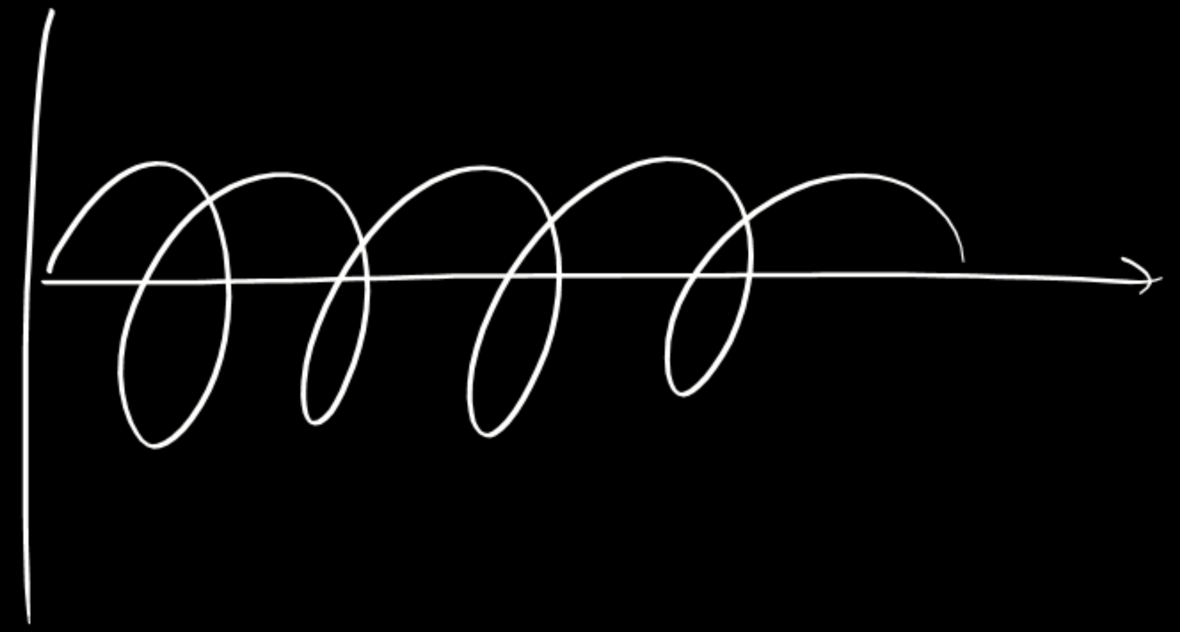
(A) its z-coordinate can never be negative

(B) its x-coordinate can never be positive

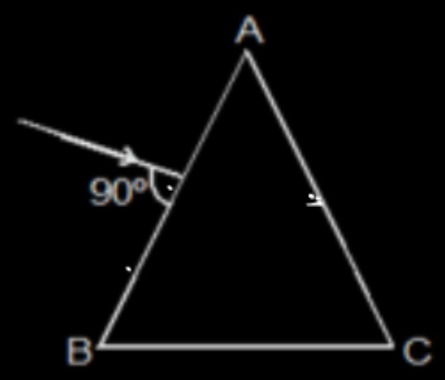
(C) its x-and z-coordinates cannot be zero at the same time

(D) its y-coordinate will be proportional to its time of flight





A ray of light is incident normally on one of the equal faces of an isosceles glass prism. The light ray is reflected internally twice on the same sized faces and then emerges through the base of the prism perpendicularly then choose matching?

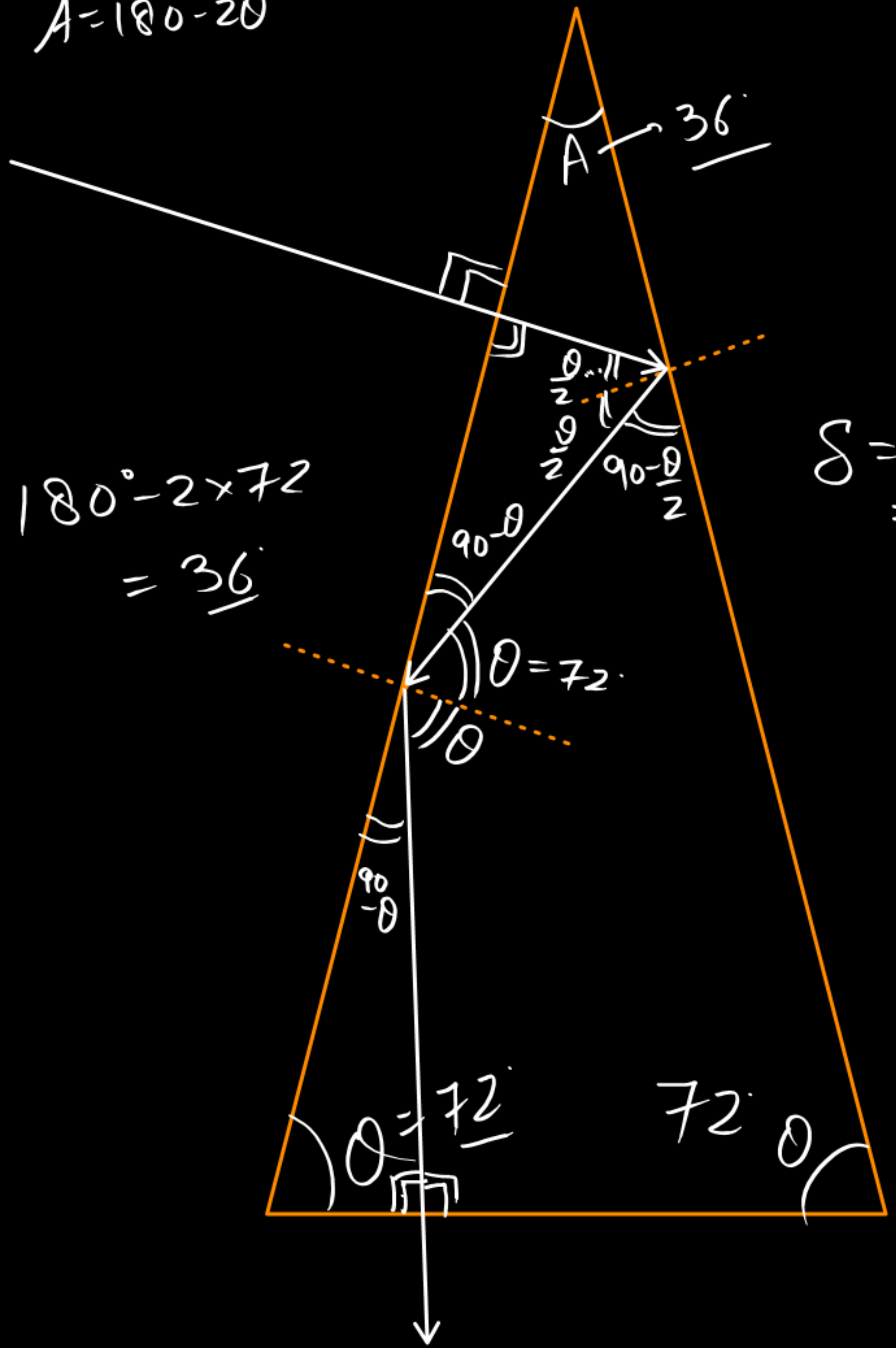


List-I	List-II
(P) Prism angle ($\angle A$)	(1) 36
(Q) $\angle B$ or $\angle C$	(2) 72
(R) Deviation at face ~~AB~~ Ac	(3) 108
(S) Deviation at face AB	(4) 144

Codes :

	P	Q	R	S
(A)	1	2	3	1
(B)	3	4	2	1
(C)	4	2	1	3
(D)	4	1	2	3

$A = 180° - 2\theta$



$180° - 2 \times 72$

$= 36°$

$\delta = 180° - 2 \times 36$

$= 108°$

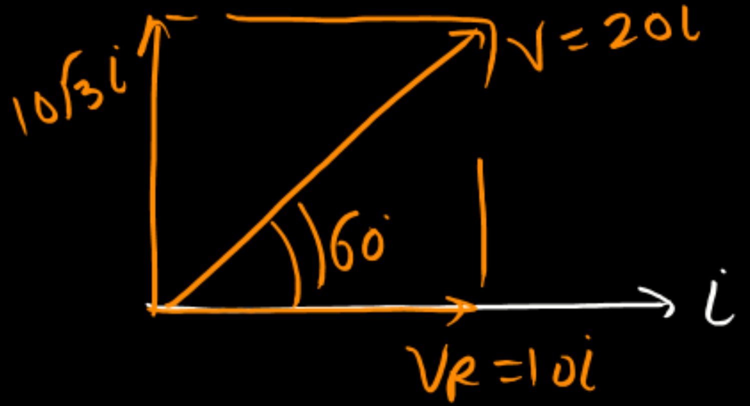
$\cancel{90} + \theta + \cancel{90} - \frac{\theta}{2} + 2\theta = \cancel{360}$ 180

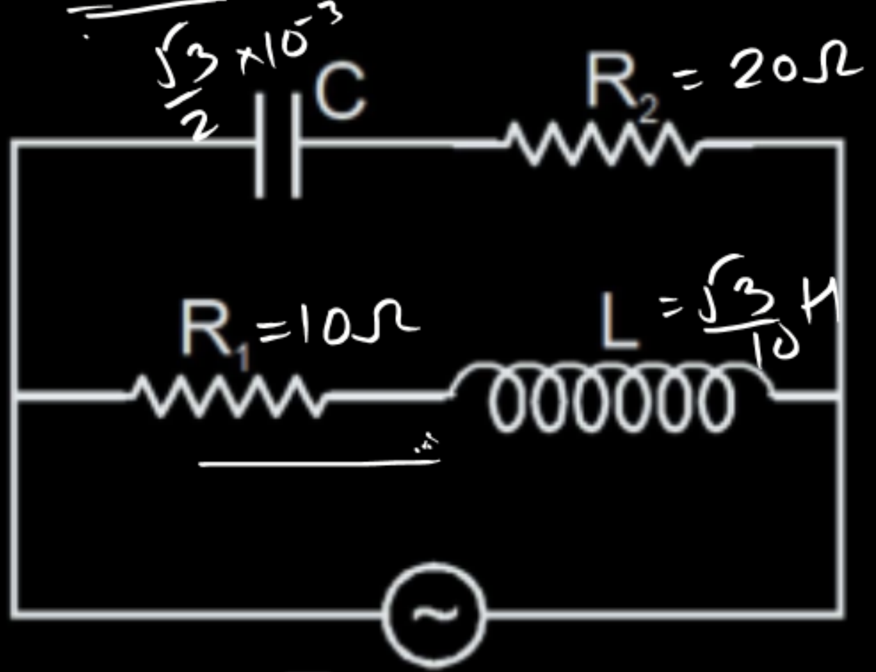
$2.5\theta = 180$

$\boxed{\theta = 72°}$

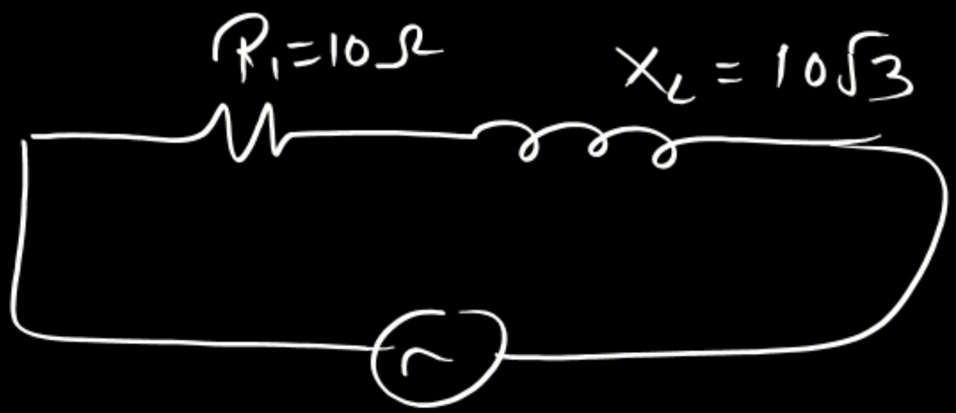
In the shown circuit, $R_1 = 10\Omega$, $L = \frac{\sqrt3}{10}$ H, $R_2 = 20\ \Omega$, $C = \frac{\sqrt3}{2}$ milli-farad and t is time in seconds. Then at the instant current through R_1 is $10\sqrt2$ A ; find the current through resistor R_2 in amperes.

0





V=$200\sqrt2$ sin(100t) volt

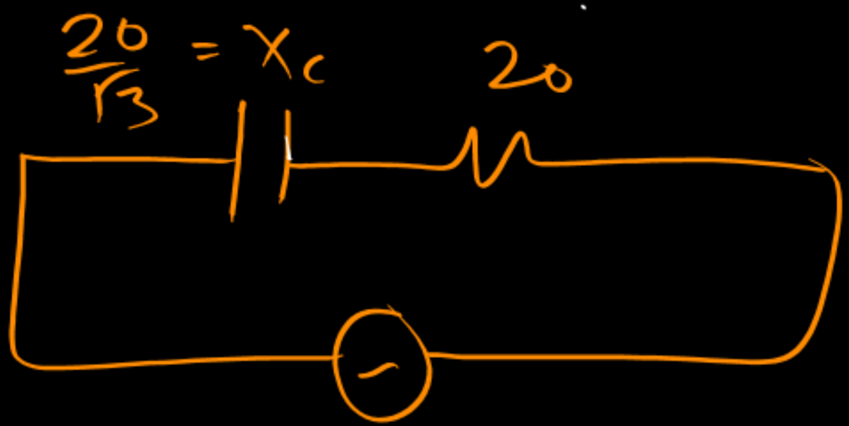


$200\sqrt2 \sin 100t$

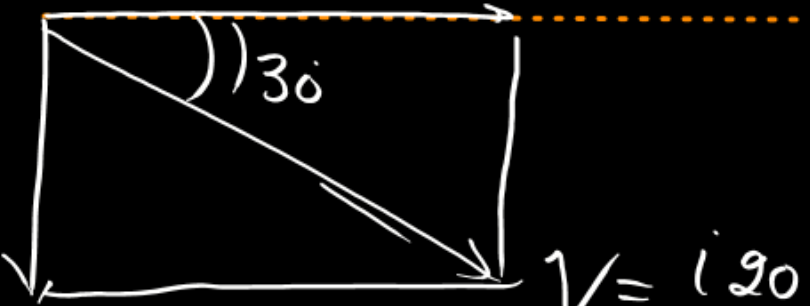
$i_{max} = \frac{200\sqrt2}{20} = 10\sqrt2$

$\boxed{i = 10\sqrt2 \sin(100t - 60)}$

$X_C = \frac{2}{100\times\sqrt3\times10^{-1}} = \frac{20}{\sqrt3}$



$20\sqrt{1+\frac13} = 20\times\frac{2}{\sqrt3}$



$\frac{20i}{\sqrt3}$

$i_2 = \frac{200\sqrt2\sqrt3}{40}\sin(100t+30)$

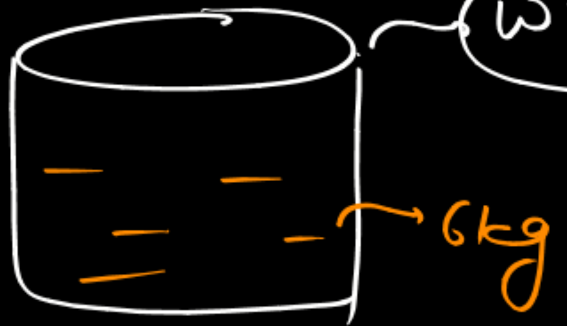
$V = i\,20\sqrt{1+\frac13} = 20i\times\frac{2}{\sqrt3} = \frac{40i}{\sqrt3}$

A vessel contains 6 kg water at 0°C. Water equivalent of vessel at 0°C is 2 kg and it increases linearly with temperature such that it becomes 4 kg at 100°C. The temperature of the system is raised from 0°C to 100°C. If Q_1 is the energy consumed by water and Q_2 is the energy consumed by vessel then $\frac{Q_1}{Q_2}$ is

(Assume all relations in °C)

0°C $\quad$ 100°C

$W = 2\,kg \rightarrow 4\,kg$



$$Q_1 = m S \Delta T$$

$$= 6\,\cancel{kg} \times 1 \frac{kcal}{\cancel{kg}\,\cancel{°C}} \times 100\,\cancel{°C} = 600\,kcal$$

(2)

$$W = 2 + C\,T$$

$$4 = 2 + C \times 100$$

$$C = \frac{2}{100}$$

$$\boxed{W = 2 + \frac{2}{100} \cdot T}$$

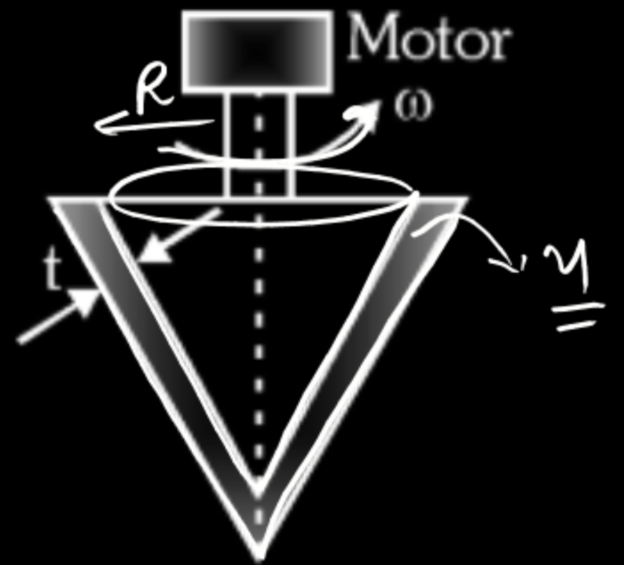
$T \rightarrow dT$

$$\int_0^{100} \left(2 + \frac{2}{100} T\right) \times 1 \times dT = 2 \times 1 \times \int_0^{100} dT + \frac{2}{100} \times 1 \int_0^{100} T\,dT$$

$$= 2 \times 1 \times 100 + \frac{\cancel{2}}{\cancel{100}} \times 1 \times \frac{\cancel{100} \times 100}{\cancel{2}}$$

$$= 200 + 100 = 300$$

The shown figure, there is a conical shaft rotating on a bearing of very small clearance t. The space between the conical shaft and the bearing, is filled with a viscous fluid having coefficient of viscosity $\eta$the shaft is having radius R and height h. If the external torque applied by the motor is x and the power delivered by the motor is P working in 100% efficiency to rotate the shaft with constant ω then



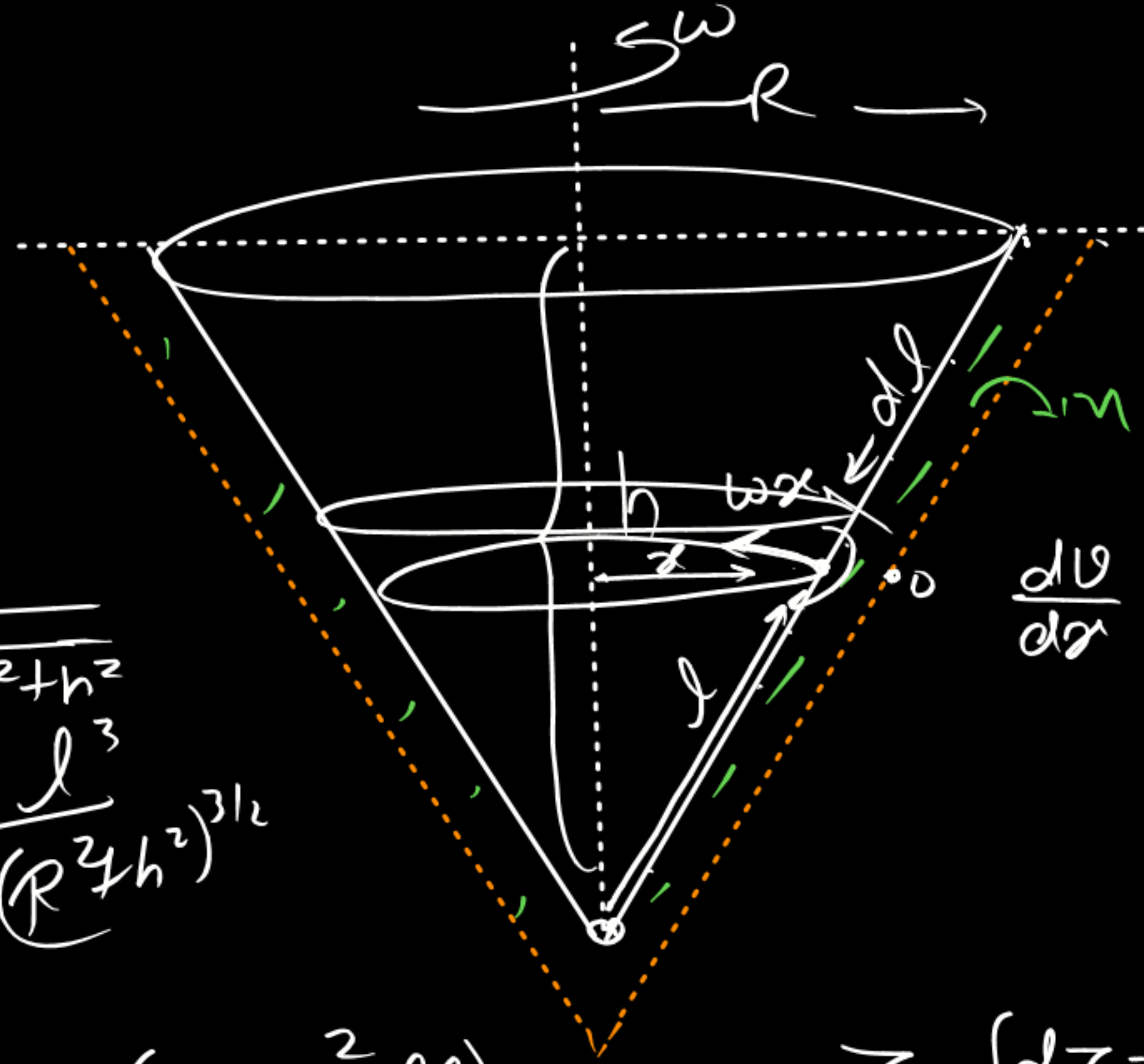
$$P = \frac{dW}{dt} = \tau\frac{d\theta}{dt} = \tau\omega$$

(A) $P = \dfrac{\pi\omega^2\eta R^3\sqrt{R^2+h^2}}{2t}$

(B) $\tau = \dfrac{\pi\omega\eta R^3\sqrt{R^2+h^2}}{2t}$

(C) $P = \dfrac{\pi\omega^2\eta R^2 h}{2t}$

(B) $\tau = \dfrac{\pi\omega\eta^2 R^2 h^2}{2t\sqrt{R^2+h^2}}$

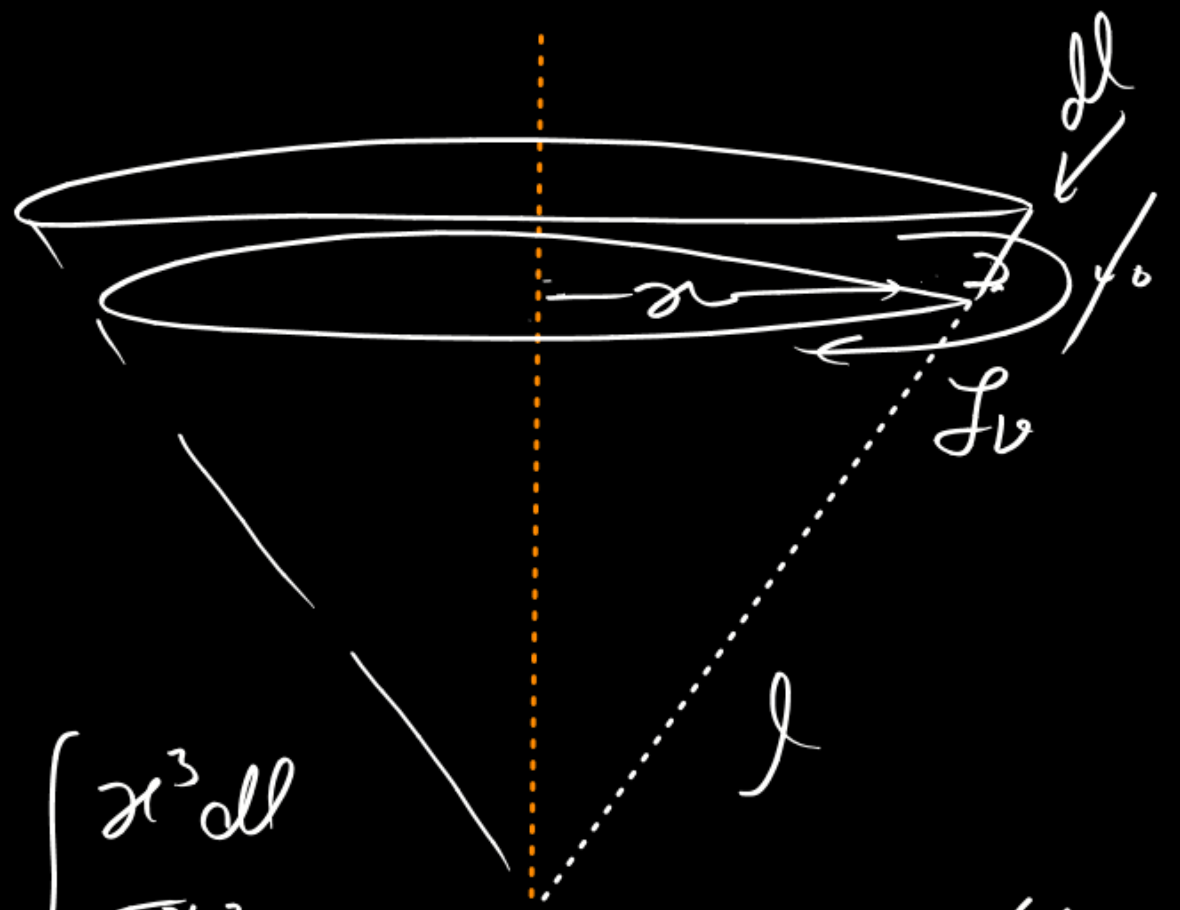


$$\frac{x}{l} = \frac{R}{\sqrt{R^2+h^2}}$$

$$x^3 = \frac{R^3\, l^3}{(R^2+h^2)^{3/2}}$$

$$\frac{dv}{dx} = \frac{\omega x}{t}$$

$$f_v = \eta A \frac{dv}{dx}$$



$$d\tau = \eta(2\pi x^2 dl)\,\frac{\omega x}{t}$$

$$\tau = \int d\tau = \frac{\eta 2\pi\omega}{t}\int_0^{\sqrt{R^2+h^2}} x^3\, dl$$

$$\tau = \frac{2\pi\eta\omega}{t}\,\frac{R^3}{(R^2+h^2)^{3/2}}\int_0 l^3\, dl = \frac{2\pi\eta\omega R^3}{t\,(R^2+h^2)^{3/2}}\cdot\left(\frac{(R^2+h^2)^2}{4}\right)$$

$$\boxed{\tau = \frac{\pi\omega\eta R^3\sqrt{R^2+h^2}}{2t}}$$

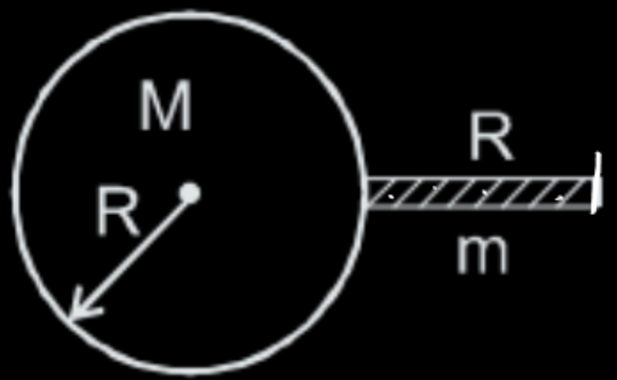
A uniform thin rod of mass m and length R is placed normally on surface of earth as shown. The mass of earth is M and its radius is R. If the ratio of magnitude of gravitational force exerted by earth on the rod and gravitational potential energy of the rod is $\frac{1}{\eta R \ell n 2}$ then 'η' is

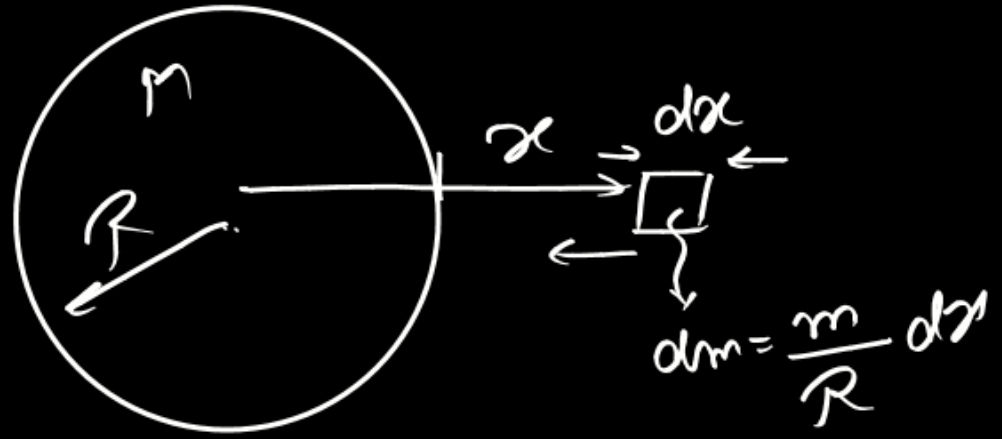
$$-\frac{GM\cancel{m}\ \cancel{R}}{2R^2\, G\cancel{Mm}\ln 2} = -\frac{1}{2R\ln 2}$$

$$\boxed{\eta = -2}$$

$$f = \frac{GMm}{R}\left(\frac{1}{R} - \frac{1}{2R}\right)$$

$$= \frac{GMm}{2R^2}$$





$$dU = -\frac{GM}{x}\frac{m}{R}dx$$

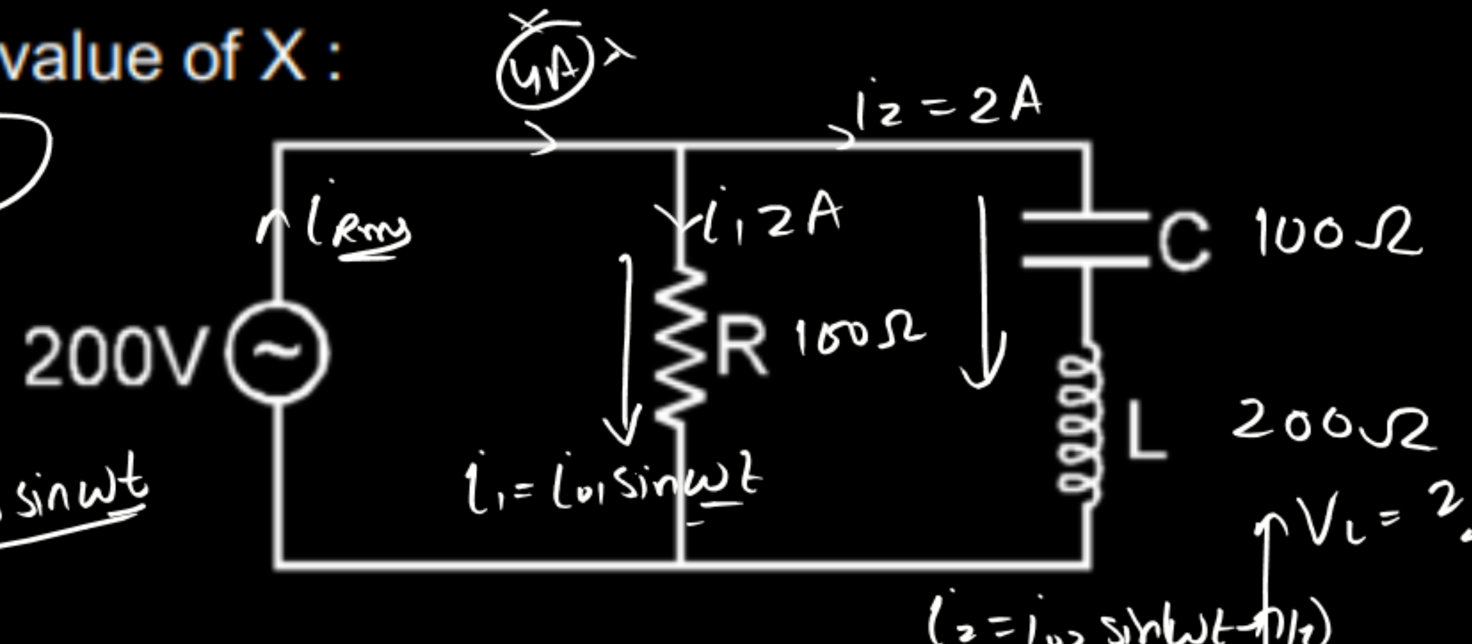
$$U = -\frac{GMm}{R}\int_R^{2R}\frac{dx}{x} = -\frac{GMm}{R}\ln 2$$

$$df = \frac{m}{R}dx\,\frac{GM}{x^2}$$

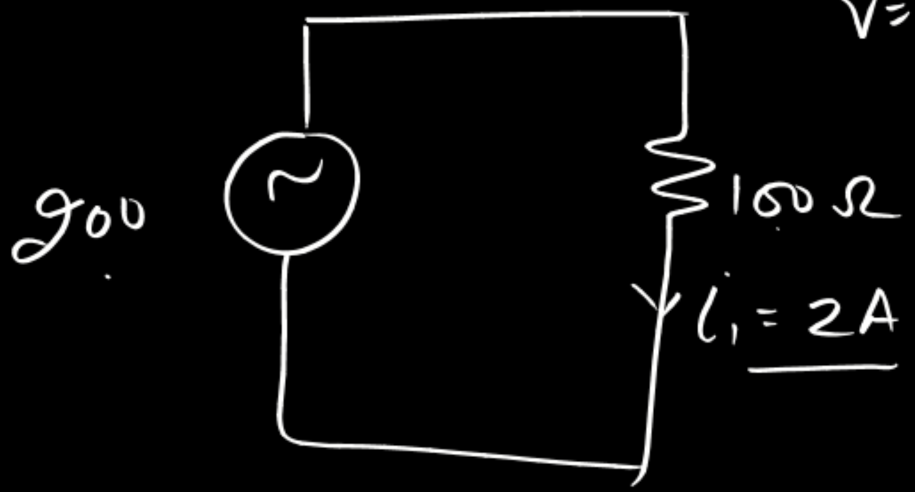
$$f = \frac{GMm}{R}\int_R^{2R}\frac{dx}{x^2} = \frac{GMm}{R}\left(\frac{-1}{R}\right)$$

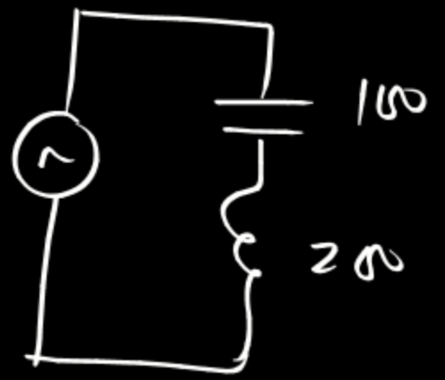
In the circuit diagram shown, $X_C = 100\ \Omega$, $X_L = 200\ \Omega$ & $R = 100\ \Omega$. The effective current through the source is $\sqrt{X}$ then find out value of X :

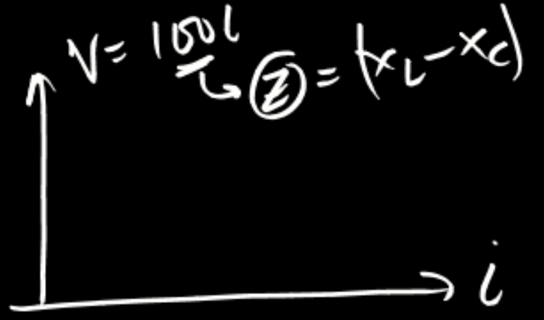
$\sqrt{8}$ $\boxed{x = 8}$



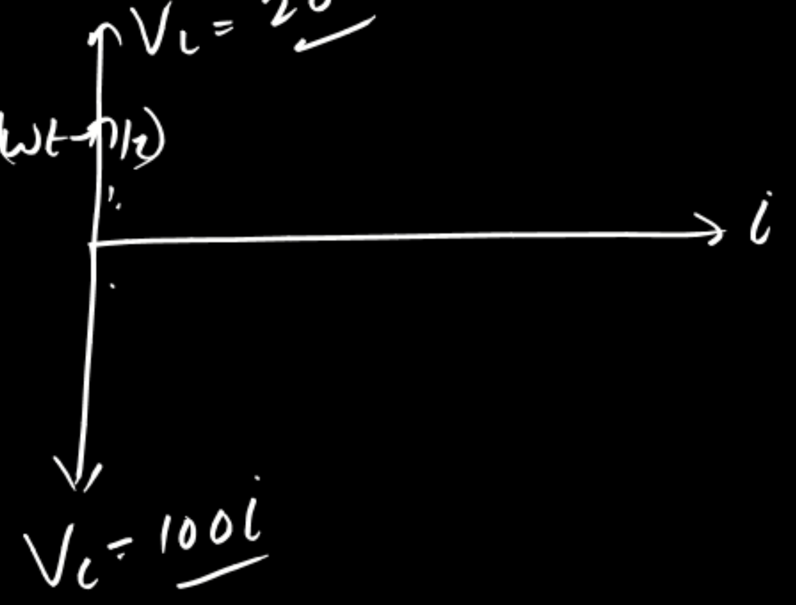
$V = V_0 \sin\omega t$







$i_2 = \frac{V}{Z} = \frac{200}{100} = 2A$

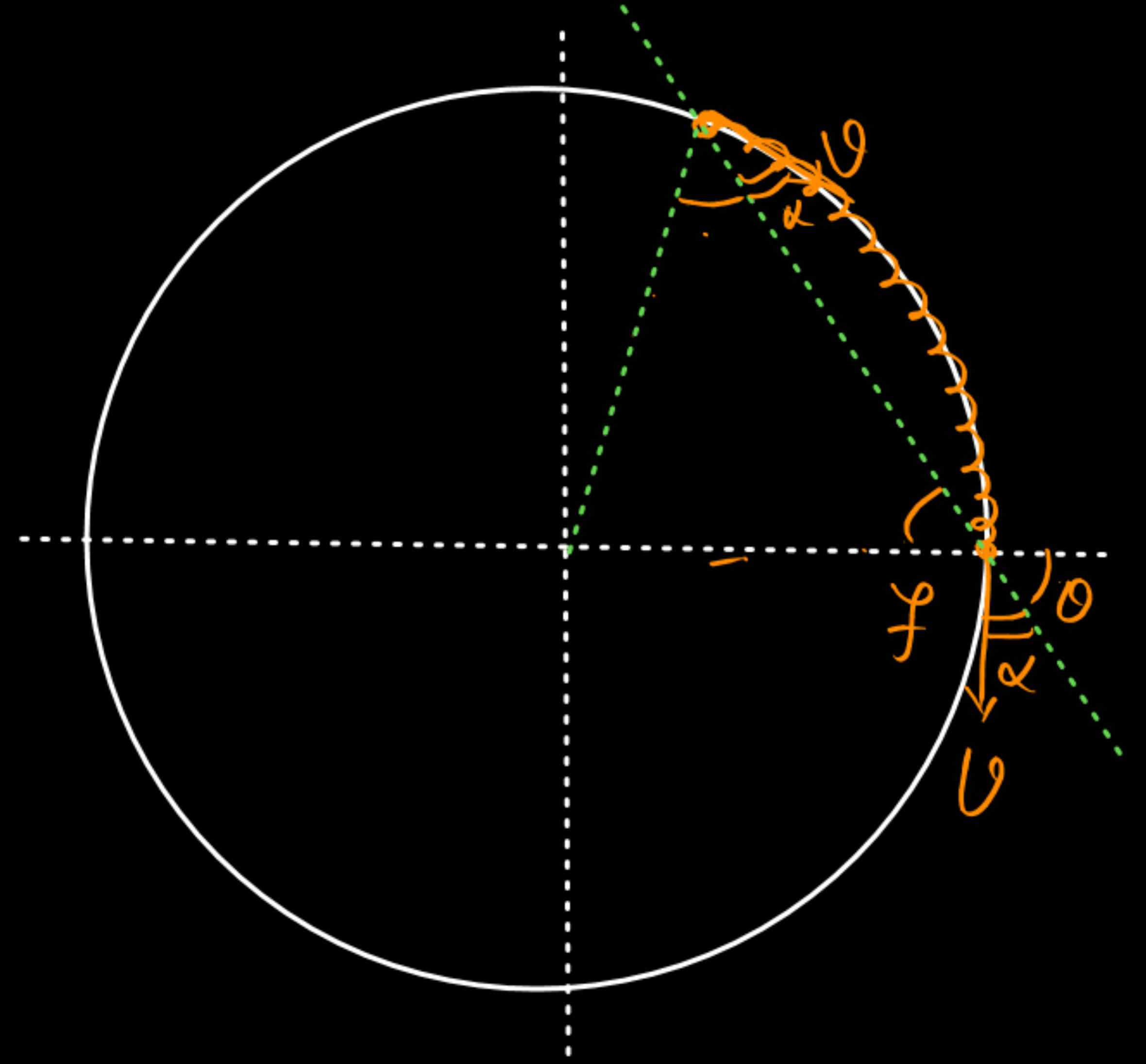


$i = \sqrt{2^2 + 2^2 + 2\times2\times2\cos\pi/2}$

$= 2\sqrt{2} = \sqrt{8}$

A train of length ℓ is moving with a constant speed v along a circular track of radius R. The engine of the train emits a sound of frequency f. The frequency heard by a guard at the rear end of the train is $f'\left(\frac{x}{y}\right)f$. The value of $(x + y)$ is (Assume that the length of train is less than $\left(\frac{\pi R}{2}\right)$.

$f' = \frac{x}{y} f$



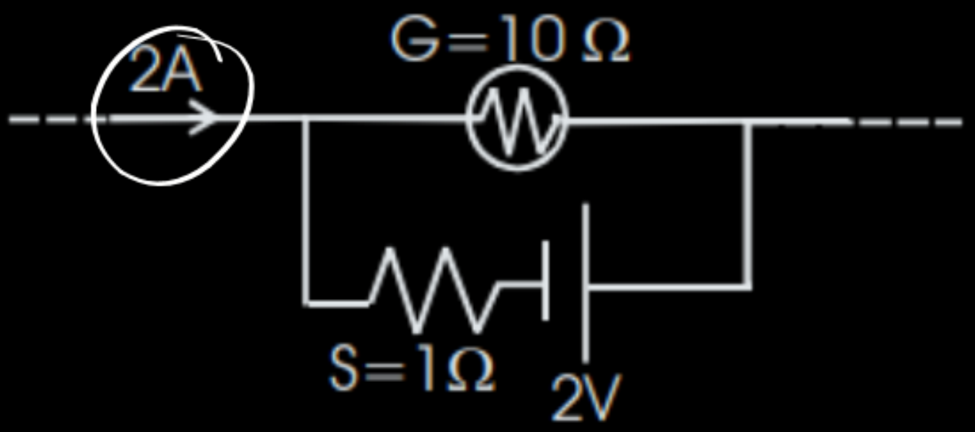
$$f' = \frac{c + v\cos\alpha}{c + v\cos\alpha} f$$

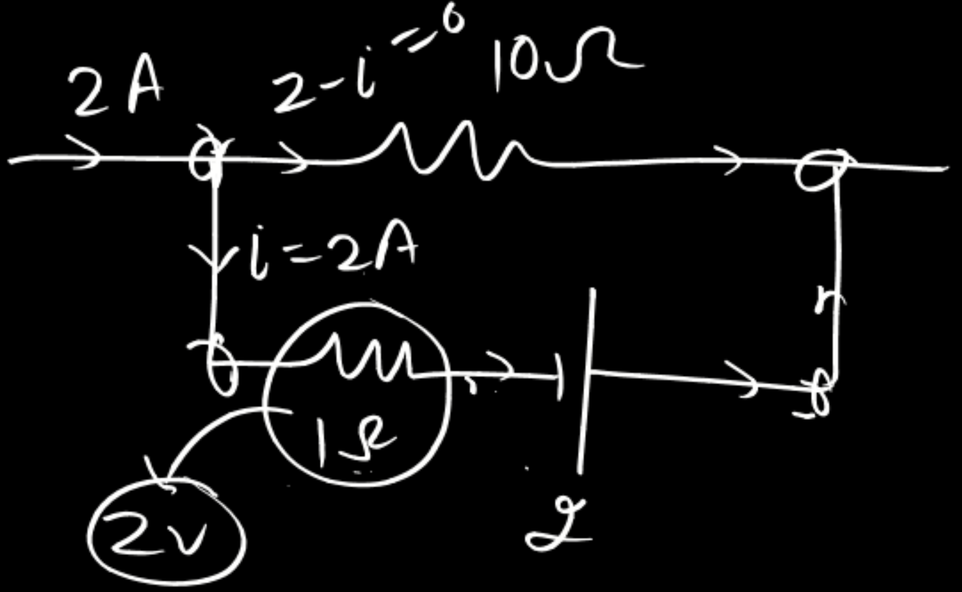
$$= \frac{1}{1} f$$

$$\boxed{x + y = 2}$$

The galvanometer shown in the figure has resistance 10Ω. It is shunted by a series combination of a resistance $S = 1\Omega$ and an ideal cell of emf 2V. A current 2A passes as shown. The potential difference (in volts) across the resistance S is

2





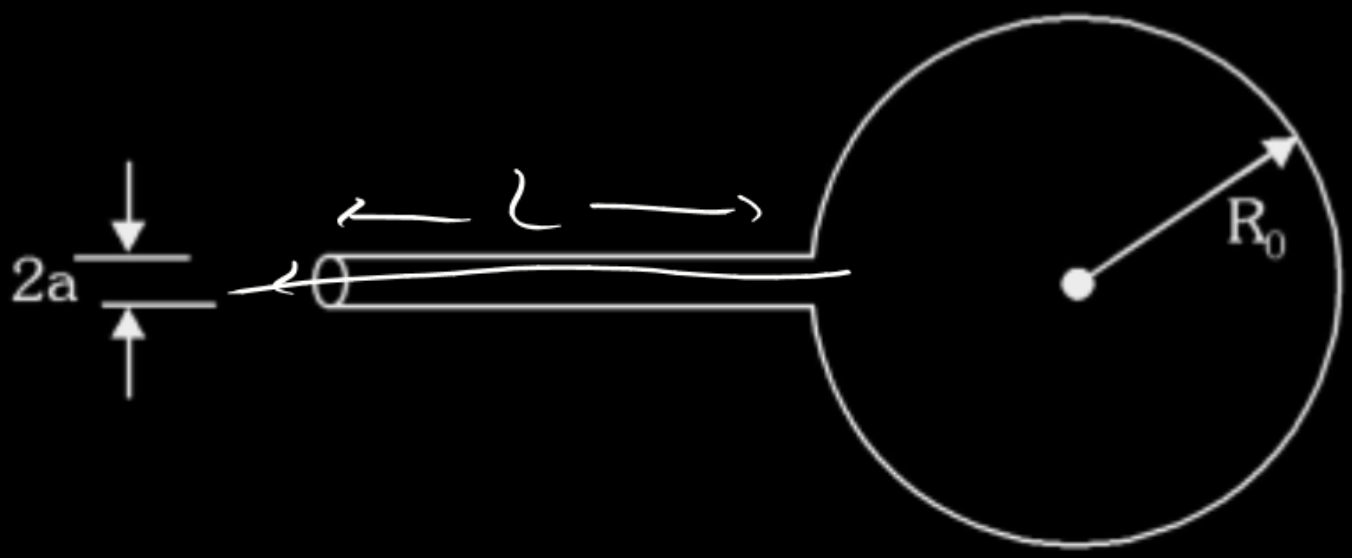
$$-i+2+10(2-i)=0$$

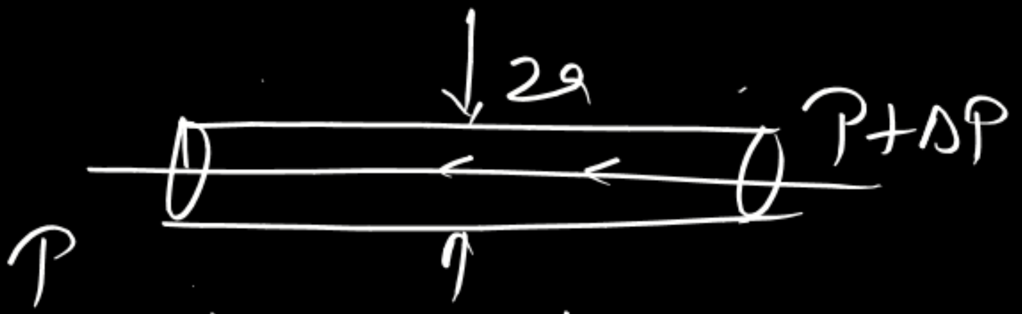
$$-i+2+20-10i=0,$$

$$22=11i$$

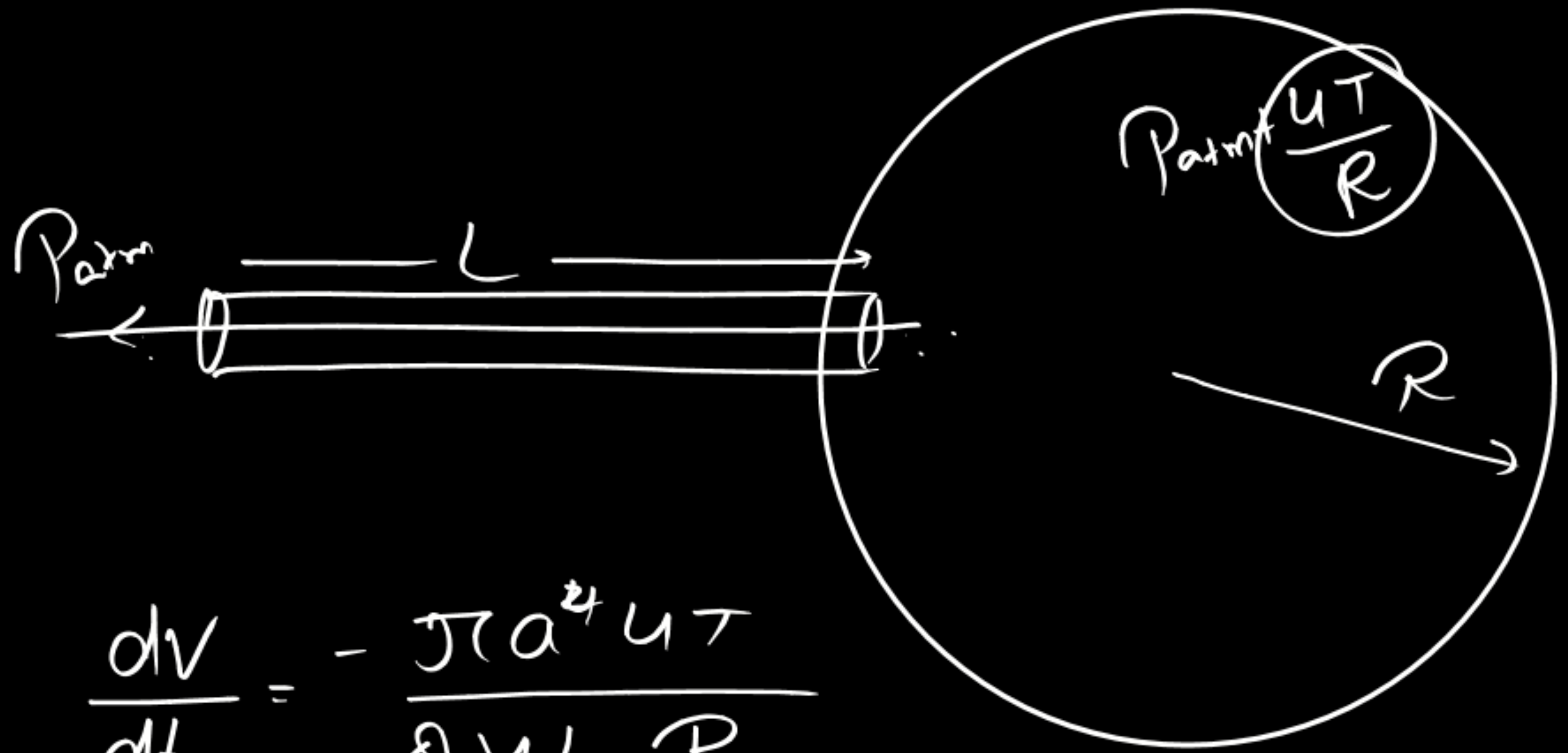
$$i=2A$$

A soap bubble is blown at the end of a capillary tube of radius a and length L. When the other end is left open, the bubble begins to deflate. The radius of the bubble as a function of time is given by $R_0\left[1-\dfrac{a^4Tt}{m\eta LR_0^4}\right]^{\frac{1}{n}}$ (where initial radius of the bubble was R_0. Surface tension of soap solution is T. It is known that volume flow rate through a tube of radius a and length L is given by Poiseuille's equation – $Q=\dfrac{\pi a^4\Delta P}{8\eta L}$; ΔP is pressure difference at the two ends of the tube and r\ is coefficient of viscosity. Assume that the bubble remains spherical) Find the value of m × n.





$$\frac{dV}{dt}=\frac{\pi a^4\Delta P}{8\eta L}$$



P_{atm}

$$V = \frac{4}{3}\pi R^3$$

$$dV = \frac{4}{3}\pi\, 3R^2\, dR$$

$$-\frac{a^4 T}{8\eta L}\int_0^t dt = \int_{R_0}^{R} R^3\, dR$$

$$-\frac{a^4 T}{2\cancel{8}\eta L}\, t = \frac{R^4 - R_0^4}{\cancel{4}}$$

$$\frac{dV}{dt} = -\frac{\pi a^4\, 4T}{8\eta L\, R}$$

$$= -\frac{\cancel{\pi} a^4 T}{2\eta L R} = \frac{4\cancel{\pi} R^2\, dR}{dt}$$

$$R^4 = \left(R_0^4 - \frac{a^4 T}{2\eta L}\, t\right) \Rightarrow R = R_0\left(1 - \frac{a^4 T t}{R_0^4\, 2\eta L}\right)^{1/4}$$

$m \times n$

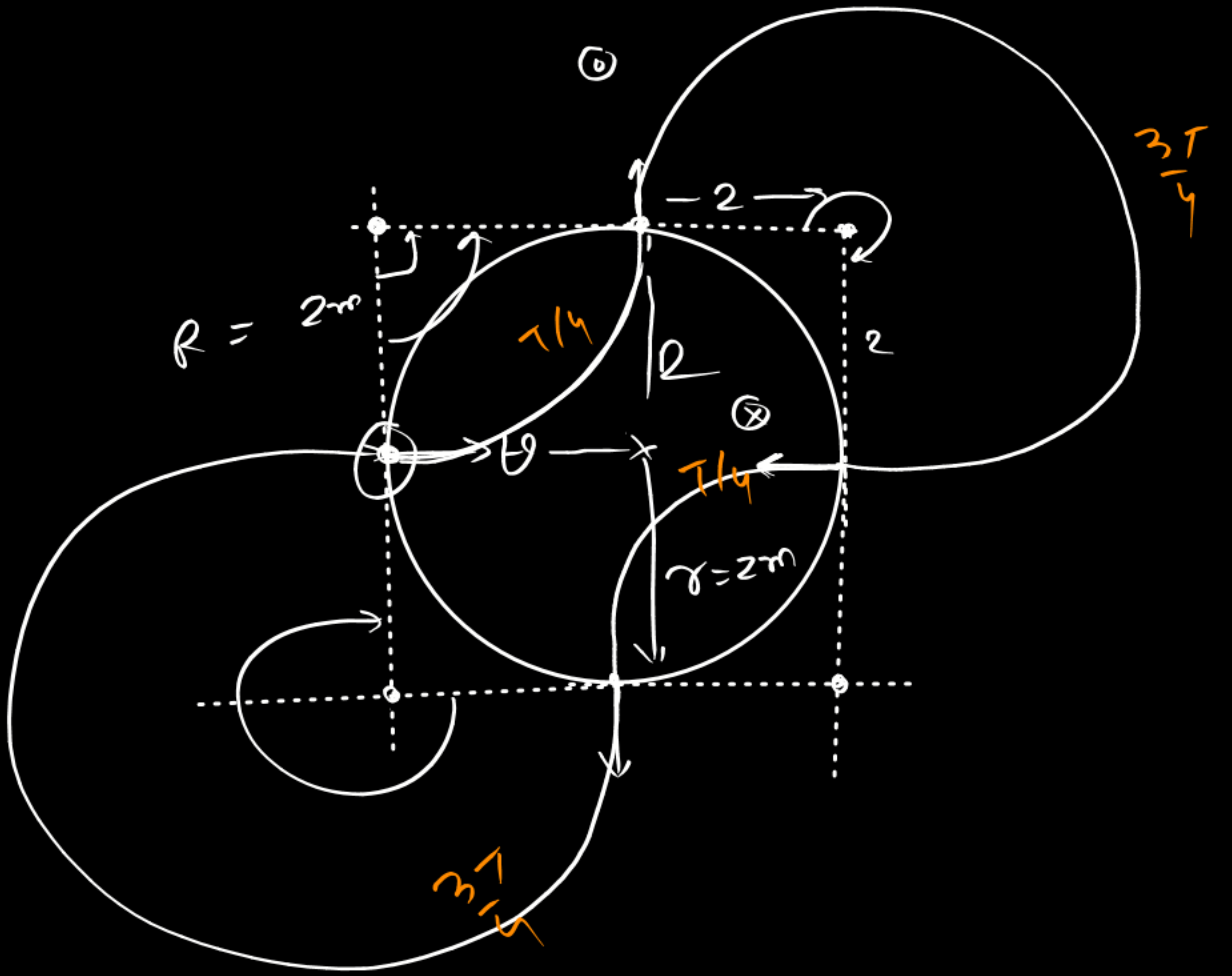
$$\boxed{2 \times 4 = 8}$$

There exist a uniform inward magnetic field B =1T in cylindrical region of radius r = 2 m. Outside the cylindrical region the magnetic field is outward but of same magnitude 1T. A point charge (q = 1C and m = 1 kg) is projected from a point A on circumference with velocity 2m/s towards center as shown in figure. After time t = $N\pi$ sec particle again passes through initial point A. Find the minimum possible value of N.

4π

$N = 4$

$$R = \frac{1 \times 2}{1 \times 1}$$
$$= 2$$



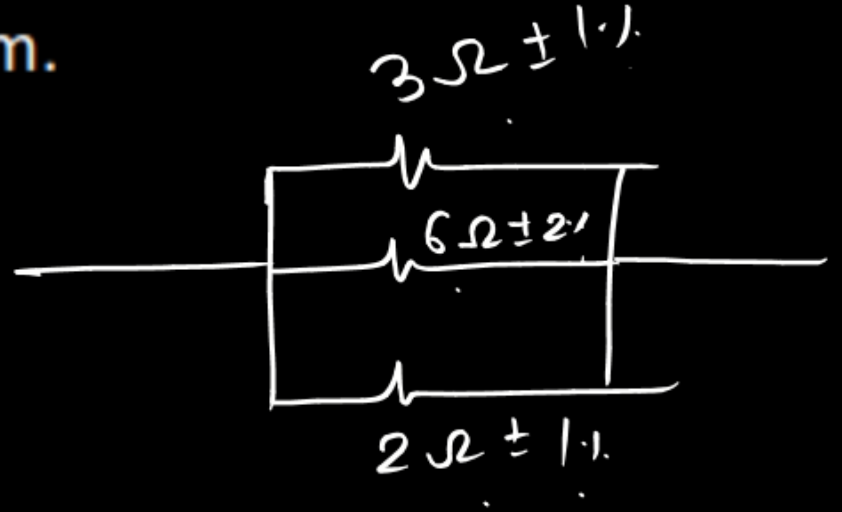
$$t = 2T = 2 \times \frac{2\pi m}{qB}$$
$$= \frac{4\pi \times 1}{1 \times 1} = 4\pi \text{ sec}$$

Three resistances are measured in Ohm.

$R_1 = 3\Omega \pm 1\%$

$R_2 = 6\Omega \pm 2\%$

$R_3 = 2\Omega \pm 1\%$



$$\frac{1}{R_{eq}} = \frac{1}{3} + \frac{1}{6} + \frac{1}{2} = \frac{2+1+3}{6}$$

$$\boxed{R_{eq} = 1\Omega}$$

When they are connected in parallel, maximum percentage error in equivalent resistance is $\frac{\beta}{6}$. Here β is an integer. Find β

$$\frac{\beta}{6} = \frac{7}{6} \qquad \beta = 7$$

$$\frac{dR_1}{R_1} \times 100 = 1$$

$$\frac{dR_2}{R_2} \times 100 = 2$$

$$\frac{dR_3}{R_3} \times 100 = 1$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{dR_{eq}}{R_{eq}^2} = \frac{dR_1}{R_1^2} + \frac{dR_2}{R_2^2} + \frac{dR_3}{R_3^2}$$

$$\left(\frac{dR_{eq}}{R_{eq}} \times 100\right) = R_{eq}\left\{\frac{dR_1}{R_1} \times 100 \times \frac{1}{R_1} + \frac{dR_2}{R_2} \times 100 \times \frac{1}{R_2} + \frac{dR_3}{R_3} \times 100 \times \frac{1}{R_3}\right\}$$

$$= 1\left\{1 \times \frac{1}{3} + 2 \times \frac{1}{6} + 1 \times \frac{1}{2}\right\} = \frac{1}{3} + \frac{1}{3} + \frac{1}{2} = \frac{2+2+3}{6} = \frac{7}{6}\%$$

Power factor of an L-R circuit is 0.6 and that of a C-R circuit is $\sqrt{\frac{36}{37}}$. If the elements [L, C and R] of the two circuits are joined in series the power factor of the circuit is found to be 1. The ratio of resistance in the C-R circuit to the resistance in the L-R circuit is m : 1. Find the value of m.

$\boxed{X_L = X_C}$ $\qquad \frac{R_2}{R_1}$

$\cos\theta = \frac{R}{Z} = \frac{R_1}{\sqrt{R_1^2+X_L^2}} = \frac{6}{10}$

$\frac{R_1^2+X_L^2}{R_1^2} = \frac{100}{36}$

$\frac{R_2^2+X_C^2}{R_2^2} = \frac{37}{36}$

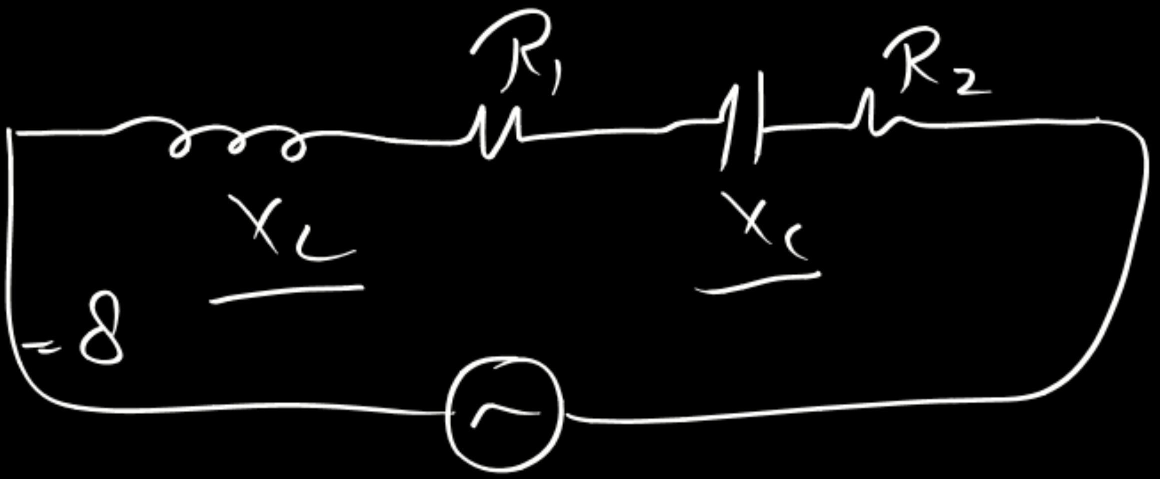
$1+\frac{X_C^2}{R_2^2} = \frac{37}{36}$

$\cos\theta = \frac{R_2}{\sqrt{R_2^2+X_C^2}} = \sqrt{\frac{36}{37}}$

$1+\frac{X_L^2}{R_1^2} = \frac{100}{36}$

$\frac{X_L^2}{R_1^2} = \frac{64}{36}$

$\frac{X_C}{R_2} = \frac{1}{6}$



(1) $\cos\theta = \frac{R}{Z} = \frac{R_1+R_2}{\sqrt{(R_1+R_2)^2+(X_L-X_C)^2}} = 1$

$\boxed{\frac{X_L}{R_1} = \frac{8}{6}}$

$\frac{X_L/R_1}{X_C/R_2} = \frac{8/6}{1/6} = 8$

$R_2/R_1 = 8:1$